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Diabetic retinopathy screening using machine learning: a systematic review

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Abstract

Diabetic retinopathy (DR) stands as a leading cause of global blindness. Early identification and prompt treatment are crucial in preventing vision impairment caused by diabetic retinopathy (DR). Manual screening of retinal fundus images is challenging and time-consuming. Additionally, there is a significant gap between the number of DR patients and the number of medical experts. Integrating machine learning (ML) and deep learning (DL) techniques is becoming a viable alternative to traditional DR screening techniques. However, the absence of a retinal dataset with standardized quality, the complexity of DL models, and the need for high computational resources are challenges. Therefore, in this study, we studied and analyzed the research landscape in integrating ML techniques in DR screening. In this regard, our work contributes significantly in several aspects. Initially, we identify and characterize images of the retinal fundus that are readily available. Then, we discuss commonly used preprocessing techniques in DR screening. In addition, we analyze the progress of ML techniques in DR screening. Lastly, we discussed existing challenges and showed future directions.

Keywords Deep learning, Diabetic retinopathy screening, Machine learning, Transfer learning, Computer vision

Introduction

Diabetic retinopathy is a medical condition in which damage occurs to the retina due to diabetes. According to Antonetti et al. [1], DR is the main cause of blindness in people between the ages of 20 and 74. More than half of all individuals with diabetes (49.7%) were projected to be not diagnosed, according to statistics [2]. Globally, it is estimated that 34.6% of diabetic patients have DR, with 10.2% of those individuals experiencing visual impairment [3].

As a result, DR is a potentially dangerous eye condition that affects people and requires early diagnosis. Currently, retinal fundus images are manually scanned by ophthalmologists to determine if DR is present in patients and to grade the severity of the disease. According to the study [4], there were around 232, 866 ophthalmologists worldwide, with a significant disparity between low-income (3.7 ophthalmologists per million patients) and high-income countries (76.2 ophthalmologists per million patients). Additionally, in Sub-Saharan African countries 2.5 ophthalmologists per million people [5]. This highlights the gap between the limited number of ophthalmologists and the millions affected by DR. Additionally, manually screening retinal fundus images for DR diagnosis is time-consuming and tedious. Patients with DR, therefore, do not receive timely medical attention.

Nevertheless, the American Academy of Ophthalmology advises that individuals with type 1 diabetes should

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have their first eye exam five years after the disease first appears, and patients with type 2 diabetes should have an eye exam at the time of diagnosis [6]. However, the majority of diabetic individuals wait until their retinopathy gets severe enough to affect their ability to see clearly before undergoing a DR examination.

DR is a progressive eye disease characterized by various retinal lesions, which indicate the severity of the condition [7]. These lesions can be broadly classified into microaneurysms (MA), hemorrhages (HM), and soft and hard exudates (EX). Microaneurysms, the earliest signs of DR, appear as tiny, round red spots on the retina, typically smaller than 125 micrometers with well-defined boundaries, resulting from weakened vessel walls [8]. Hemorrhages, in contrast, are larger red patches with irregular edges and diameters exceeding 125 micrometers, categorized into flame-shaped (superficial) and blot-shaped (deeper) forms [9]. Hard exudates, caused by blood leakage, manifest as yellow patches with clear borders, whereas soft exudates, resulting from nerve fiber swelling, appear as white ovals [10]. Hemorrhages and microaneurysms are collectively referred to as red lesions, while exudates form white lesions [8].

The clinical stages of DR range from mild, moderate, and severe non-proliferative DR to proliferative DR, with or without macular edema [11]. Ophthalmologists rely on these lesions for screening, localization, classification, grading, and monitoring of disease progression. Figure 1 illustrates these lesions, enhancing their visual interpretation [12].

Recent advancements in AI and DL systems have significantly enhanced DR diagnosis, offering precise, efficient, and cost-effective solutions to screen individuals with diabetes while supporting accurate severity-level prediction [13]. This systematic review aims to analyze the existing literature, focusing on studies that have significantly contributed to the development of automated methods for DR screening. It highlights both the strengths and limitations of machine learning techniques, particularly in terms of computational complexity. This comprehensive review is intended to provide valuable insights for researchers looking to explore and advance the field of DR screening. The main contributions of our review are as follows:

- Identifying the retinal fundus imaging datasets most commonly used in research.
- Discussing the types of artifacts found in retinal fundus images and the most commonly used preprocessing techniques.
- Reviewing studies on DR grading, localization, and segmentation.
- Highlighting the top-performing AI models used for DR screening.
- Identifying the challenges in the field and suggesting future research directions.

The remaining sections of this paper are organized as follows. In Sect. 2, we have explored the related works associated with the systematic review of DR. Sect. 3 is dedicated to discussing the methodology employed in our research. Additionally, Sect. 4 provides a comprehensive list of available open datasets for DR screening, and in Sect. 5, we discussed mostly utilized performance measuring metrics. In Sect. 6, we have discussed the preprocessing techniques employed for image analysis of the retinal fundus in the reviewed articles. Section 7 focuses on the screening for DR and reviews the existing literature on binary classification, grading, localization, and segmentation. Section 8 discusses the model complexity

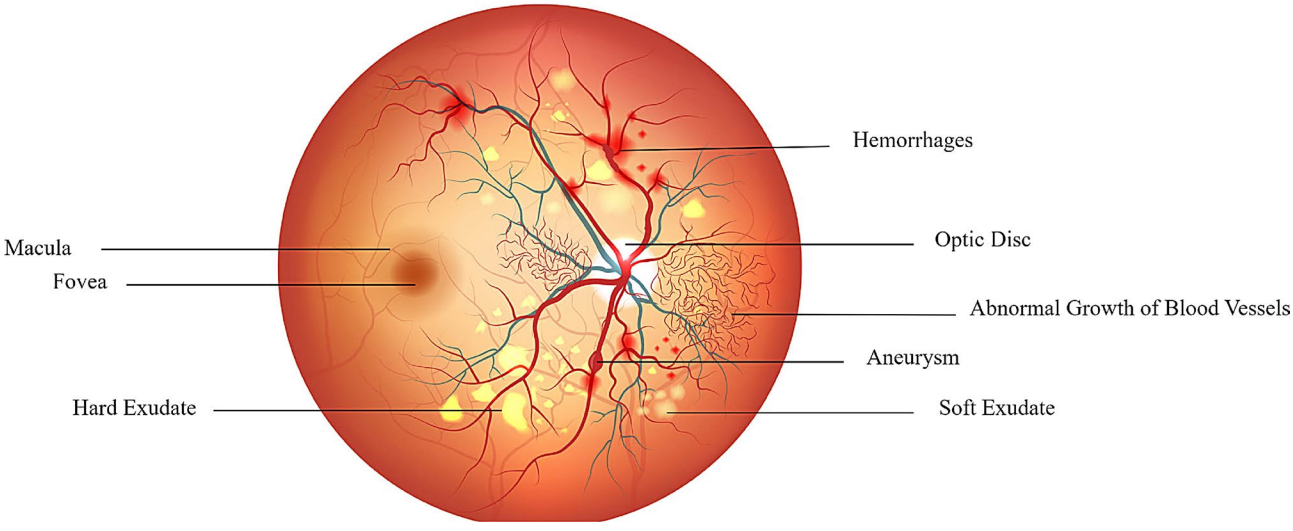


Fig. 1 Visualization of unhealthy retina with their respective lesion [12]

of reviewed studies. Section 9 discusses the current trends of AI in DR. In Sect. 10, we thoroughly examined and analyzed existing papers that utilize various techniques in this domain. Section 11 is dedicated to discussing the current challenges in DR diagnosis and future directions. Finally, Sect. 12 presents the results of our research findings and conclusions drawn from the study.

Related works

The quest to find better DR screening techniques that can help physicians in DR has been an active research area. As a result, several systematic reviews have been completed to foster research in the field and recap techniques used in DR screening.

Ishtiaq, et al. [14] reviewed papers focused on DR grading and classifications. They identified the research gaps and future directions for research communities. Additionally, they provided an analysis of the datasets, preprocessing techniques employed, and comprehensive narrations on ML models used in the research papers. Lakshminarayanan, et al. [15], articulated the publicly available dataset and accessed the requested dataset. Furthermore, they quantified the performance of ML models. In their review, Nadeem, et al. [16] have reviewed the development of DL models in the domain of DR segmentation, detection, and classification. Although Tumi Karason, et al. [17] assessed the efficacy of DR grading using the seven standard fields, questioning its significance and optimality. They explored whether comparable results could be attained through alternative methods, including the smaller field of view (FOV) or ultra-wide-field imaging.

Islam, et al. [18] have performed a systematic review to assess the performance of DL models for automated detection of DR in the retinal fundus image. On the other hand, Ademola, et al. [19] have reviewed literature in which their main objective is to build a model that is used to segment and classify DR and provide valuable information, limitations, observations, and future work. Although Sebastian et al. [20] quantified the performance of DL models and identified the best-performing models.

Additionally, Tsiknakis et al. [21] cover various DL approaches for DR detection and classification, including analyses of model performance. However, the review does not discuss the computational complexity of each paper. Selvachandran et al. [22] comprehensively cover research articles in the domain of DR, providing in-depth analyses of models like CNN and ANN. The review also discusses AI trends in the field of DR but lacks a comparative analysis of the models. In contrast, Alyoubi et al. [23] provide a thorough overview of current research articles, including a detailed examination of available datasets for DR and identification of challenges in the research area. However, they do not discuss model complexity or the resource intensiveness of the models. Furthermore, Chu et al. [24] review only 15 research articles, focusing on a few prominent algorithms. Despite identifying barriers, the paper lacks sufficient actionable recommendations. Grzybowski et al. [25] provides an overview of various AI technologies for DR, including several systems used, but it lacks a comparison and in-depth analysis of the methodologies employed in each research article. On the other hand, Wewetzer et al. [26] reviewed only 10 research articles, whereas Wu et al. [27] demonstrates significant strengths, including a comprehensive review of 60 studies and adherence to systematic methodologies. However, it fails to discuss the computational complexity of each proposed model in the research articles in greater depth.

Our systematic review bridges the gaps in prior review articles by delivering a comprehensive analysis and comparison of recent studies related to DR. Unlike previous works, our systematic review discusses the computational complexity of the reviewed studies in terms of model parameter count, providing insights into their resource intensiveness. Additionally, we identify key challenges, such as dataset limitations, artifacts, preprocessing techniques, and lack of model robustness, while offering actionable recommendations to address these issues.

Table 1 provides a summary of related systematic reviews, highlighting their respective contributions and limitations.

Methodology

This section outlines the method used to review DR screening. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines are employed to define the research questions, databases, and search terms necessary for conducting the systematic review. PRISMA serves as a framework for the literature review process [28].

Research questions

In this systematic review work, we have defined four main questions with their respective aim. These questions were made to be precise, concise, and focused on the review's particular goals to provide a guide for the methodical gathering and examination of data. The research questions and their main objectives or motivations are summarized in Table 2.

Search strategies

The selection of the database and search strings is tailored to address the research questions highlighted in Sect. 3.1. The search primarily focused on the identification of relevant literature from key databases such as

Table 1 Summary of related systematic reviews, including their contributions and limitations

Author and Year	Contributions	Limitations
Ishtiaq, et al., 2020 [14]	• Provides analysis of the datasets and preprocessing techniques employed. • Provides comprehensive narrations on ML models used in the literature.	• The review paper has a narrow discussion.
Alyoubi et al., 2020 [23]	• Provides a thorough overview of a current research article. • Includes detailed examination of available datasets for DR.	• Failed to address the computational complexity of the model.
Mohaimenul Islam, et al., 2020 [18]	• Quantifies the DL algorithms' performance in detecting DR and points out the best-performing DL.	• They have assessed only 23 literature within the span of 19 years.
Chu et al., 2020 [24]	• Reviewed literature on DL algorithms tackling barriers to clinical implementation: generalization, limited screening scope, and data sovereignty.	• Only 15 studies are reviewed.
Grzybowski et al., 2020 [25]	• Extensively reviewed various AI technologies. • Highlighted the strength of articles and areas for improvement.	• Limited focus on the implementation of the proposed AI model in the article.
Lakshminarayanan, et al., 2021 [15]	• Describes the publicly available dataset and access on request dataset. • Quantifies the performance of ML models.	• The review paper lacks a discussion on the ML models employed in the research papers.
Tsiknakis et al., 2021 [21]	• It covers various DL approaches used for DR detection and classification. • It analyzes the performance of the model.	• Failed to address the computational complexity of the model.
Wewetzer et al., 2021 [26]	• The inclusion of various diagnostic parameters allows for naunced understanding of the performance of DL models. • The use of quality assessment to evaluate the quality of included studies.	• Only 10 studies are included.
Wu et al., 2021 [27]	• Demonstrated ML algorithms exhibit high sensitivity and specificity in diagnosing DR. • It is a comprehensive review that includes 60 studies.	• Failed to delve into the computational complexity of the models proposed om each study.
Tumi Karason, et al., 2022 [17]	• Determines whether seven standard fields are the most optimal screening methods for DR grading.	• The study is limited to analyzing the impact of retinal imaging on the DL model.
Selvachandran et al., 2023 [22]	• Comprehensive coverage of research articles. • In-depth analysis of models like CNN, ANN, etc. • Discuss the trends of AI in the field of DR.	• Research article selection bias. • Lack of comparative analysis of the model.
Sebastian, et al., 2023 [20]	• Lists of commonly used public datasets in the screening of DR. • Quantifies the performance of DL models and points out the best-performing models.	• Lacks a clear comparison of models employed..
Ademola, et al., 2023 [19]	• Provides an analysis of the CNN methods utilized to classify and segment retinal fundus images.	• Minimal attention given to literature centered around classification.

Table 2 Summary of research questions and objectives

No.	Research Questions	Aims
RQ1	Are there any freely available retinal fundusdataset? What are their characteristics?	Identify DR datasets and their features.
RQ2	What are the common pre-processing techniques used for data cleaning or denoising, data transformation?	Identify insightful retinal fundus image pre-processing mechanisms that enhance the performance of the ML models
RQ3	What methods are utilized in detecting, grading, and segmenting DR?	Distinguish and discuss existing DR grading, detection, and segmentation techniques.
RQ4	What challenges are associated with the techniques used for DR identification?	Identify restrictions, challenges, and future directions.

Google Scholar, PubMed, IEEEXplore, ScienceDirect, MDPI, and Scopus.

In this review, we conducted a comprehensive literature search using specific search strings across multiple databases to gather relevant studies related to the application of ML and DL techniques in DR. The search strings were carefully designed to target studies focused on the detection, classification, screening, segmentation, and grading of DR, with an emphasis on imaging techniques such as funduscopic and ophthalmoscopic methods. The strings used in Google Scholar and Scopus database are the following: [("Machine Learning" OR "Deep Learning" OR "CNN" OR "Convolutional Neural Network" OR "Neural Network") AND "Diabetic Retinopathy" AND ("Detection" OR "Classification" OR "Screening" OR "Segmentation" OR "Grading") AND ("Funduscopic" OR "Ophthalmoscopic")], for PubMed, IEEE Xplore, and MDPI is: [("DNN" OR "CNN") AND "Diabetic Retinopathy" AND ("Detection" OR "Classification" OR "Screening" OR "Segmentation" OR "Grading")], for ScienceDirect is: [("Machine Learning" OR "Deep Learning" OR "CNN" OR "Convolutional Neural Network") AND "Diabetic Retinopathy" AND ("Funduscopic" OR "Ophthalmoscopic")], for ACM Digital Library is: [("Machine Learning" OR "Deep Learning" OR "CNN" OR "Convolutional Neural Network" OR "Neural Network") AND "Diabetic Retinopathy" AND ("Detection" OR "Classification" OR "Screening" OR "Segmentation"

OR "Grading")]. The complete procedure for identifying relevant research papers is outlined in Algorithm 1.

Algorithm 1 Pseudocode for search query strings

```
Databases ← [Google Scholar, PubMed, IEEE Xplore, MDPI, ScienceDirect, Scopus]
Define Keywords
Disease ← "Diabetic Retinopathy"
Aim ← [Detection, Grading, Segmentation, Screening]
Method ← [Machine Learning, Deep Learning, Convolutional Neural Network, Neural Network]
Data ← [Funduscopic Image, Ophthalmoscopic Image]
Search_String ← " "
for method ∈ Method do
    for aim ∈ Aim do
        for data ∈ Data do
            Search_String ← method AND Disease AND aim AND data
            for database ∈ Databases do
                papers ← database.search(Search_String)
            end for
        end for
    end for
end for
```

Table 3 outlines the specific conditions or requirements that were applied during the selection process to determine which literature was included in the review and which was excluded. These criteria play a crucial role in ensuring that the selected studies align with the focus and objectives of the review. The steps involved in applying the inclusion and exclusion criteria during the selection process are represented in Fig. 2, which also shows the total number of journal articles identified for quantitative analysis. Additionally, Fig. 3 shows selected studies distribution with respect to the database. This is a clear illustration of the systematic review's literature selection process. Seventy eight (78) research papers were filtered for review after conducting an extensive search on each database and applying the inclusion and exclusion criteria.

Diabetic retinopathy datasets

DL algorithms rely extensively on large datasets to understand patterns in images that depict abnormalities and normal conditions. Training ML or DL models with substantial, high-quality datasets enhances model performance, emphasizing the significance of both image size and quality. The common retinal fundus datasets used for DR screening are APTOS 2019 [29], MESSIDOR [30], MESSIDOR 2 [31], DIARETDB0 [32], DIARETDB1 [33], DDR [34], EyePACS [35], DRIVE [36], STARE [37], CHASE DB1 [38], and IDRiD [39]. Additionally, extensive and various aspects of fundus imaging technology and its applications in healthcare as been covered [40].

The **APTOS 2019** dataset [29], available on Kaggle, includes 5,590 retinal fundus images with five severity levels (0, 1, 2, 3, 4) to detect DR.

The **MESSIDOR** dataset [30] contains 1,200 color posterior pole eye fundus photos, captured using a Topcon TRC NW6 non-mydriatic retinograph and a 3CCD camera across three ophthalmologic departments. The photos, with 8 bits per color plane, include 800 taken with pupil dilation and 400 without, primarily for DR grading

The **DIARETDB0/1** dataset [32, 33] contains images captured with various digital fundus cameras with a 50-degree FOV. The camera settings and photometric information are unknown, leading to potential variance in image quality due to factors like noise, distortion, and chromatic aberrations, making it challenging to standardize retinopathy findings.

The **DDR** dataset [34] contains 13,673 color fundus images collected from 147 hospitals across 23 provinces in China. The images, captured using 42 different fundus camera models with a 45-degree FOV, include popular models like Topcon D700, Topcon TRC NW48, Nikon D5200, and Canon CR2. All images have been desensitized to ensure privacy and utility.

The **EyePACS** [35] dataset includes pictures taken with different camera models, which leads to differences in how left and right are presented visually. There are five possible ratings for each image: 0, 1, 2, 3, and 4. A few pictures show the retina anatomically (the optic nerve in the right eye is shown on the right, and the macula on the left). Conversely, some replicate an inverted

Table 3 Criteria for selecting relevant research papers

No.	Inclusion	Exclusion	Description
C1	x	-	Studies published between 2020-2024
C2	x	-	Studies that are either conference or article
C3	x	-	Studies on DR detection, grading, and segmentation using fundus image
C4	-	x	Studies written in other languages except English
C5	-	x	Duplicated studies
C6	-	x	Studies that do not have clear result presentation

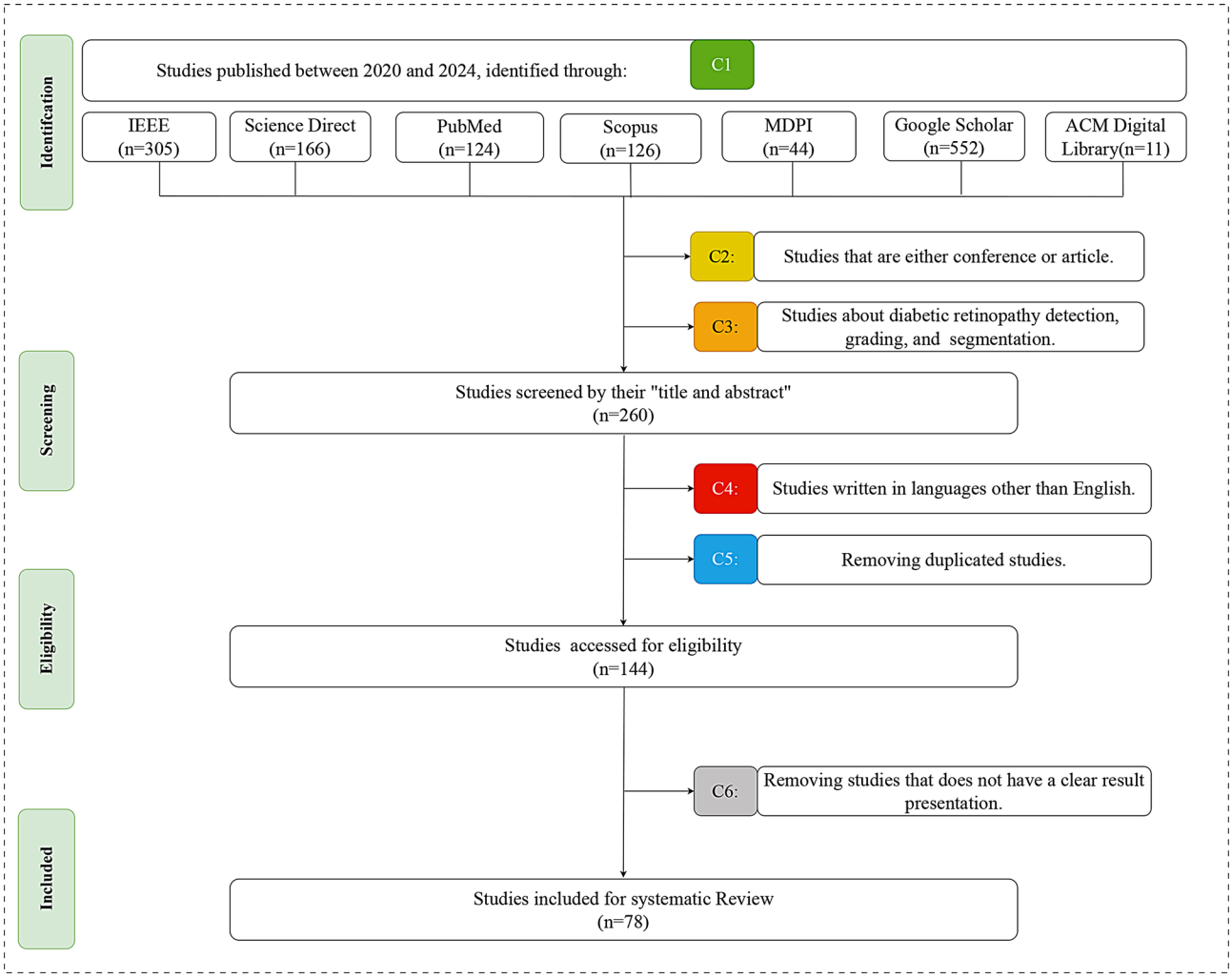


Fig. 2 Flow diagram illustrating the research paper selection process

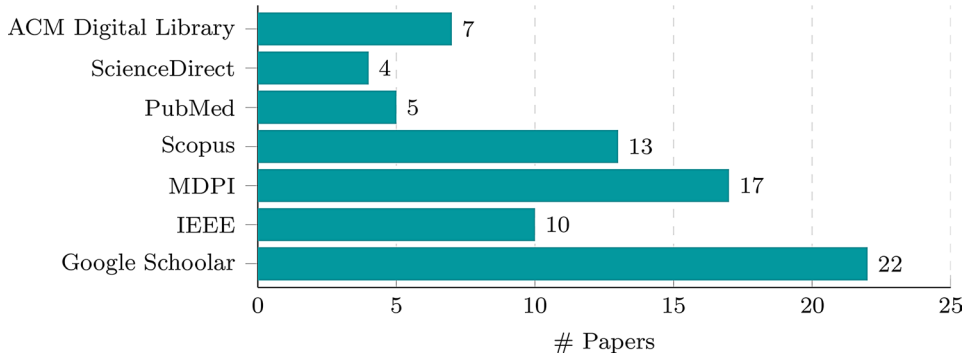


Fig. 3 Distribution of reviewed research papers across databases

microscope-condensing lens image, which is similar to a live ocular examination.

The **DRIVE** dataset [36] consists of 40 retinal fundus images, 7 of which show abnormal pathology, captured using a Canon CR5 3CCD camera with a 45-degree FOV. Each image has a resolution of 584×565 pixels and 8 bits per color channel. The dataset is divided into a training set and a testing set, each with 20 images. The training set includes a circular FOV mask and expert manual segmentation, while the testing set features two manual segmentations, with the first serving as the benchmark for performance evaluation.

A **STARE** [37] dataset for segmenting retinal vessels is called STARE (Structured Analysis of the Retina). It includes 20 retinal fundus images with equal size (700× 605 pixels).

The **CHASE_DB1** dataset [38] is a retinal vascular segmentation dataset consisting of 28 color retina images, each with a resolution of 999×960 pixels. These images are taken from the left and right eyes of fourteen school-age children. Each image has been annotated by two independent domain experts.

The **IDRiD** dataset [39] contains 516 fundus images, categorized into normal retinal images and those showing signs of diabetic macular edema (DME) or DR. It includes ground truth data for DR and DME labels, fovea center and optic disc coordinates, pixel-level annotations of lesions, and disease severity grading.

The **MAPLES-DR** dataset [41] includes 198 fundus images from the MESSIDOR database, enhanced with pixel-wise annotations for 10 anatomical and pathological biomarkers related to DR and ME. It aims to improve the explainability of screening models by providing detailed segmentation maps and diagnostic labels.

The **E-Ophtha dataset** [42] serves as a benchmark for evaluating automated methods for detecting exudates and microaneurysms. The dataset consists of two subsets: E-Ophtha-EX, which contains 47 images with exudates and 35 images without lesions, and E-Ophtha-MA, which includes 148 images of microaneurysms or small hemorrhages and 233 images without lesions.

The **ROC** Dataset [43] was developed for the Retinopathy Online Challenge to identify microaneurysms in retinal images. It includes 100 images, split into a training set and a testing set of 50 images each. The training set provides reference locations for microaneurysms. Images were captured using Topcon NW 100, Topcon NW 200, or Canon CR5-45 NM cameras and saved in JPEG format.

Table 4 summarizes the most commonly used public datasets, showing the total number of samples, the resolution of the retinal fundus images, and their respective categories.

Performance metrics

Performance metrics assess a model's effectiveness, efficiency, and overall performance in classification and segmentation tasks.

- **True Positive (TP)**: Correctly identified positive instances or segmented abnormalities.
- **False Positive (FP)**: Incorrectly identified positive instances or wrongly segmented abnormalities.
- **False Negative (FN)**: Missed positive instances or unsegmented abnormalities.
- **True Negative (TN)**: Correctly identified negative instances or background regions.

The following are the performance metrics used in DR classification and segmentation;

Accuracy: it measures how accurate the model is in predicting the right output from a given input as defined in Eq. (1).

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{1}$$

Precision: the precision of a model is calculated by dividing the total number of true positive predictions by the total number of positive predictions as defined in Eq. (2).

Table 4 Summary of commonly used public datasets for diabetic retinopathy

No.	Dataset	Samples (DR: Normal)	Resolution (Pixels)	Category	URLs (Accessed on: May 13, 2024)
1	MESSIDOR [30]	1200 (660:540)	Variable	Grading	https://www.adcis.net/en/third-party/messidor/
2	MESSIDOR-2 [31]	1748 (528:1217)	Variable	Grading	https://www.adcis.net/en/third-party/messidor2/
3	DIARETDB0 [32]	130 (110:20)	1500×1152	Grading	http://www.it.lut.fi/project/imageret/diaretdb0
4	DIARETDB1 [33]	89 (84:5)	1500×1152	Grading	http://www.it.lut.fi/project/imageret/diaretdb1/index.html
5	DDR [34]	12522 (6256:6266)	Variable	Grading	https://github.com/nkicsl/DDR-dataset
6	IDRiD [39]	516 (168:348)	4288×2848	Segmentation and Grading	https://idrid.grand-challenge.org/
7	APTOS 2019 [29]	5590	Variable	Grading	https://www.kaggle.com/competitions/diabetic-retinopathy-detection
8	EyePACS [35]	-	Variable	Grading	https://www.kaggle.com/competitions/diabetic-retinopathy-detection
9	DRIVE [36]	40	584×565	Segmentation	https://drive.grand-challenge.org/
10	STARE [37]	20	700×605	Segmentation	https://cecas.clemson.edu/%20ahoover/stare/
11	CHASE_DB1 [38]	28	999×960	Segmentation	https://researchdata.kingston.ac.uk/96/
12	E-Ophtha [42]	-	1440×960 to 2544×1696	Lesion Detection	https://www.adcis.net/en/third-party/e-ophtha/
13	ROC Dataset [43]	100	-	Lesion Detection	https://webeye.ophth.uiowa.edu/ROC/

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

Recall/Sensitivity: it is computed by dividing the total number of false negatives and true positives by the number of true positives and it is defined as Eq. (3).

$$Recall(Sensitivity) = \frac{TP}{TP + FN} \tag{3}$$

Specificity: it is computed by dividing the total number of false positives and true negatives by the sum of the two, as defined in Eq. (4).

$$Specificity = \frac{TN}{TN + FP} \tag{4}$$

F1 Score: an accuracy metric that combines test precision and recall is called the F1 score, and it is defined as Eq. (5).

$$F1 - Score = \frac{2 * Precision * Recall}{Precision + Recall} \tag{5}$$

AUC: the equation for the Area Under the ROC Curve (AUC) is given by Eq. (6):

$$AUC = \int_a^b F(x)d(x) \tag{6}$$

where (a) and (b) are the upper and lower limits on the curve, and F(x) is the equation of the curve.

Kappa: The Kappa metric measures the agreement between the model's predictions and the ground truth while accounting for chance agreement, as defined in Eq. (7).

$$Kappa = \frac{P_o - P_e}{1 - P_e} \tag{7}$$

where P_o is the observed agreement, and P_e is the expected agreement by chance.

Dice coefficient: it is a measure of how similar two sets are to one other and given as in Eq. (8).

$$Dice_Score(A, B) = \frac{2 \times |A \cap B|}{|A| + |B|} \tag{8}$$

where $(|A \cap B|)$ represents the size of the intersection of sets A and B, $(|A|)$ represents the size of set A, and $(|B|)$ represents the size of set B.

Jaccard Index: It is expressed as the area where two sets intersect divided by the area where the two sets join and it is defined as in Eq. (9).

$$Jaccard_Index(A, B) = \frac{|A \cap B|}{A \cup B} \tag{9}$$

where (A) and (B) are sets, $(|A \cap B|)$ is the size of the intersection of sets (A) and (B), and $(|A \cup B|)$ is the size of the union of sets (A) and (B).

Performance metrics for classification include Accuracy, Precision, Recall (Sensitivity), Specificity, F1 Score, AUC, and Kappa, which evaluate model effectiveness in classification. The Dice Coefficient and Jaccard Index are used for segmentation, measuring the similarity and overlap between predicted and ground truth regions.

Retinal fundus image artifacts and preprocessing techniques

This section discusses the most common artifacts in retinal fundus images and the pre-processing techniques employed in research papers we have reviewed. Image pre-processing involves identifying the artifacts, removing noise or variations, enhancing the quality of the retinal fundus image, and making it suitable for the model.

Artifacts in retinal fundus image

Retinal fundus images are crucial for the diagnosis of various eye diseases, including diabetic retinopathy, glaucoma, and macular degeneration. However, artifacts often degrade the quality of these images, making automated analysis more challenging. Noise can arise from several sources, such as the image acquisition process, environmental factors, or limitations of the imaging system.

A medical image is considered good if all the important structures are clearly visible [44]. In a healthy eye fundus image, the main parts that can be seen are the optic disc, macula, and blood vessels, as shown in Figure 1. There are several things that can reduce the quality of a retinal image: dust and dirt on the camera, dusty camera lenses, eye blinking, and eyelashes blocking the view [44]. Common problems caused by these factors include uneven lighting, low contrast, blurriness, and light flares, as shown in Fig. 4. Dust and dirt create light spots scattered across the image, as seen in Fig. 4a. A full eye blink results in a very poor-quality image, as shown in Fig. 4b. Incorrect gamma or flash settings can cause color issues like overexposure, underexposure, or uneven lighting, as shown in Fig. 4.

Additionally, Fundus tessellation refers to the increased visibility of large choroidal vessels due to a thinner retinal pigment epithelium (RPE) and choroid [45]. fundus tessellation itself is not noise (artifact), but if not accounted for properly, it can be challenging for AI models, resulting in misclassifications. Leopard print is shown in Fig. 5a. Furthermore, improper cleaning of the camera lenses can create circular light rings in the center of the image,

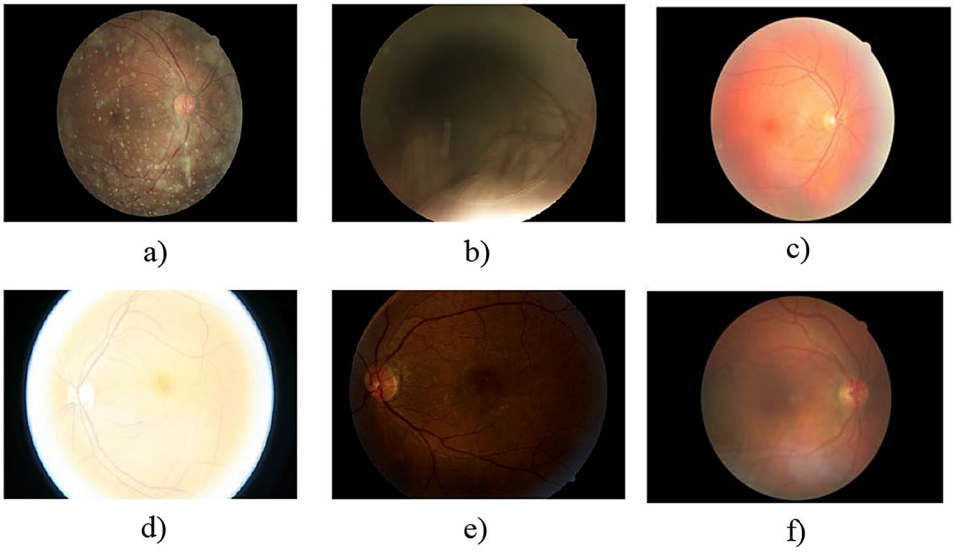


Fig. 4 Samples of low-quality fundus images with different distortions [44] **a**) Light flares, **b**) Total eye blink, **c**) Uneven illumination, **d**) Overexposed, **e**) Underexposed, **f**) Blur

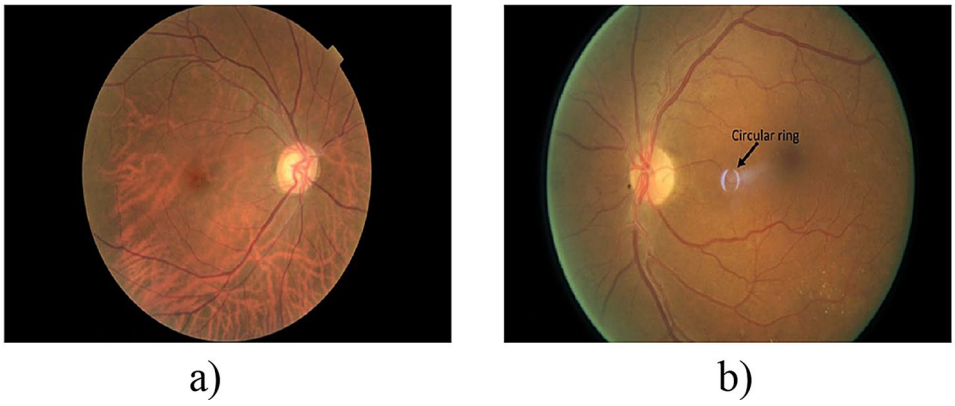


Fig. 5 Samples of fundus images with red colour 'leopard' print and circular ring [44] **a**) Leopard print, **b**) Circular ring

as shown in Fig. 5b. It is important to note that while the images in Fig. 5a and b may look like they have poor quality, experts consider them to be of acceptable quality [44].

Additionally, there are various types of noises such as Speckle noise, salt and pepper noise, Brownian noise, Gaussian noise, and Poisson noise [46]. Speckle noise is a granular noise that appears due to interference from scattered light or the imaging system's imperfections. In retinal imaging, speckle noise can distort the texture and details of the retinal layers [47]. It is especially challenging in images of the macula, where fine textures are crucial for detecting early signs of disease. Speckle noise can be mitigated using curvelet-based sigmoid mapping of histogram equalization [47] as shown in Fig. 6.

Furthermore, salt-and-pepper noise is characterized by random pixels that are either black or white, resembling salt and pepper particles [48]. This type of noise is common in image transmission processes or due to sensor defects [48]. It can obscure small structures, such as blood vessels or microaneurysms, which are critical for the diagnosis of diabetic retinopathy [49]. Salt-and-pepper noise can be removed through median filtering and CLAHE as shown in Fig. 7

Image blurring is caused by factors like motion, focus issues, lens quality, low light, optical aberrations, sensor limitations, subject movement, or atmospheric conditions [50]. This type of noise can significantly affect the accuracy of retinal image analysis, making it difficult to identify small structures or subtle changes in the retina. Blurring can be reduced using Wiener filtering and techniques proposed by the study [50]. Blurred retinal fundus image is enhanced by image decomposition, illumination correction, detail enhancement using weighted fusion, and noise suppression [50] as shown in Fig. 8.

Color and geometric transformation

In color space transformation and channel operation, techniques such as green channel extraction and grayscale

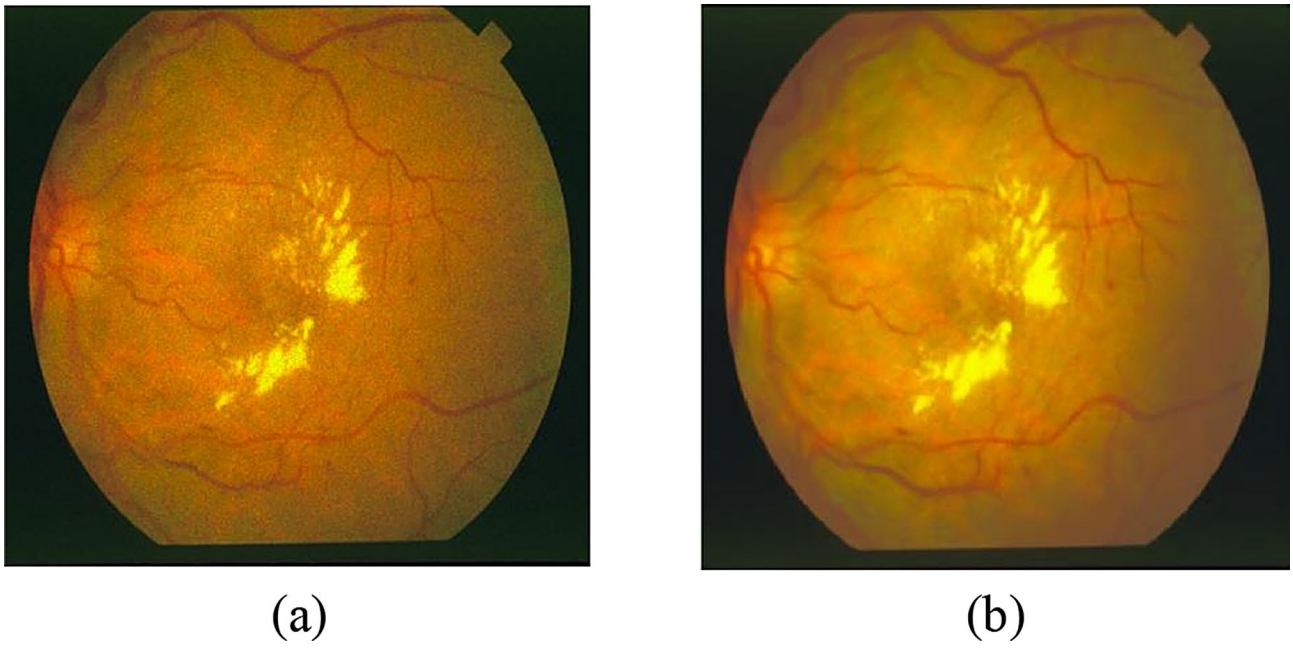


Fig. 6 Speckle noise analysis [47]; (**a**) retinal image with speckle noise, (**b**) denoised retinal fundus image with curvelet-based sigmoid mapping of histogram equalization

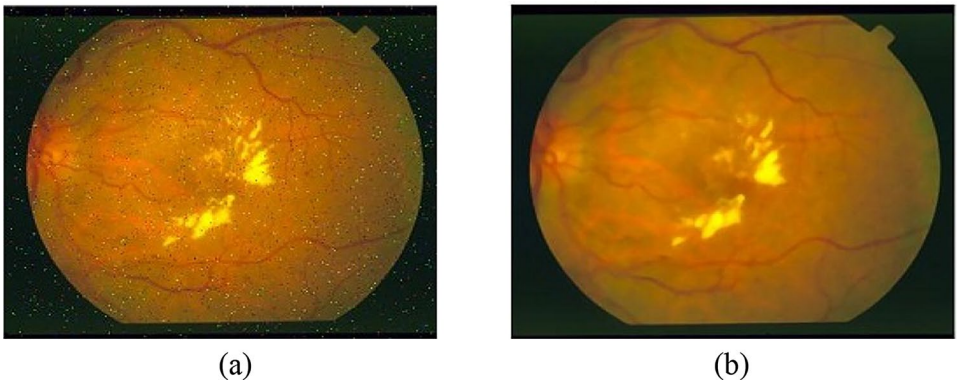


Fig. 7 Salt-pepper noise analysis [48]; (**a**) retinal fundus image with salt-pepper noise, (**b**) denoised retinal fundus image using median filter and CLAHE

conversion have been utilized in various research studies. Green channels are often used to enhance contrast or brightness and reduce noise in images of the retinal fundus [51–61]. Although grayscale conversion, which eliminates color information, is used to reduce computational overhead [62–64]. Furthermore, in the geometric transformation of retinal fundus images, cropping and resizing are two basic techniques employed in the preprocessing stage. Cropping removes irrelevant pixels from the images and is used to extract regions of interest. Resizing adjusts the size of the image to the desired resolution, ensuring compatibility with the input size of the model. Geometric transformation preprocessing techniques have been utilized in the majority of research papers we have reviewed.

Image intensity transformation and noise reduction

In image preprocessing techniques, intensity transformation is a step in which the intensity values of pixels are adjusted to improve useful features. Gamma correction, adaptive histogram equalization (AHE) [51, 65], normalization, and standardization are a few intensity-enhancing techniques employed for the image of the retinal fundus. Gamma correction is used to adjust the contrast and brightness of the image and to improve the visibility of the features in the image. On the other hand, normalization is often used to scale pixel values into the standard range [0,1], while standardization is to center and scale pixel values so that they have zero mean and unit variance. Additionally, AHE is an image processing technique that enhances local contrast by independently applying histogram equalization to small regions within



Fig. 8 Blurring noise analysis [50]; (**a**) retinal fundus image with noise, (**b**) denoised retinal fundus image

an image. Contrast Limited Adaptive Histogram Equalization (CLAHE) is an image-processing technique that extends AHE by limiting the amplification of contrast in regions with high local contrast to prevent over-enhancement and the introduction of artifacts [66]. CLAHE is one of the frontiers that has mainly utilized preprocessing techniques in the analysis of retinal fundus images [52, 54, 57, 64, 67–77].

In noise reduction techniques, Gaussian filtering [78–80], median filtering [71], and discrete wavelet transform (DWT) [67] are a few of the employed techniques in retinal fundus image analysis. Gaussian filtering is a noise reduction method in which the image is convolved with a Gaussian kernel, blurring it to reduce Gaussian noise while preserving important details [81].

Although median filtering is an image noise reduction method where each pixel's value is replaced by the median value of its neighboring pixels [82]. It's effective for removing salt-and-pepper noise.

DWT is also a powerful tool in image processing for multi-resolution analysis and denoising [83]. It is used to capture both coarse and fine-grained details effectively. It decomposes an image into approximation and detail coefficients across different scales or resolutions. The DWT operates by filtering the signal or image with a series of high-pass and low-pass filters, followed by downsampling to obtain coefficients at different scales. The process can be repeated iteratively to achieve multi-level decomposition.

Diabetic retinopathy screening

Diabetic retinopathy can be classified into five distinct stages, according to the International Clinical Diabetic Retinopathy (ICDR) [11] classification system: mild nonproliferative DR (NPDR), moderate NPDR, severe NPDR, and proliferative DR (PDR). The severity levels of DR are depicted in Fig. 9. According to the study [11], mild NPDR is characterized by the presence of MA only. Moderate DR is defined by the presence of more than just MA but not meeting the criteria for severe DR [11]. Severe DR involves more than 20 intraretinal HM in each of the four quadrants, definite venous beading in two or more quadrants, or prominent intraretinal microvascular abnormalities in at least one quadrant, with no signs of proliferative retinopathy [11]. Proliferative DR is characterized by neovascularization, abnormal blood vessel growth, or vitreous/pre-retinal hemorrhage [11].

The research carried out on DR screening uses common CNN operations. Despite utilizing the same CNN operations, studies have introduced various CNN architectures by integrating them with different supporting techniques like attention modules (residual, spatial, and channel attention) and classical ML, mainly for DR

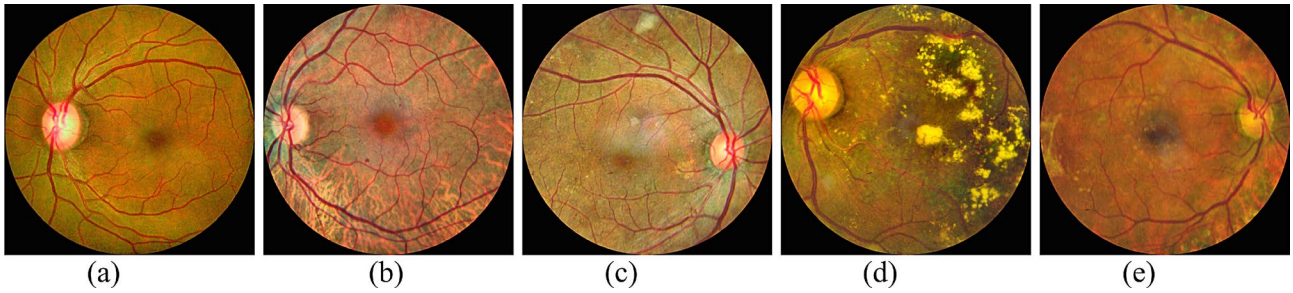


Fig. 9 Severity levels of DR [29]: (**a**) No DR, (**b**) Mild NPDR, (**c**) Moderate NPDR, (**d**) Severe NPDR, (**e**) PDR

grading. The CNN architecture depicted in Fig. 10 is a powerful tool for automated DR screening, leveraging its typical components to process retinal images effectively. The convolutional layers serve as the primary feature extraction mechanism, identifying local patterns such as edges, textures, and lesions by applying learnable filters across the image. Max pooling layers reduce the spatial dimensions of feature maps, retaining the most significant features while minimizing computational complexity, whereas min pooling captures the smallest activations, which can be useful for identifying subtle variations in features like microaneurysms. The fully connected layers integrate the extracted features to perform high-level reasoning, effectively mapping them to DR classifications. Additional components, such as activation functions like ReLU, introduce non-linearities to enable the model to capture complex patterns, and feature selection mechanisms prioritize the most relevant features for classification. This architecture systematically identifies and classifies retinal lesions, facilitating precise and efficient DR screening.

Numerous studies have been conducted to automate the screening of DR. We have classified the literature into four main categories such as binary classification, DR grading, localization, and DR segmentation.

Binary classification of diabetic retinopathy

Binary classification aims to classify input images into two categories: either as healthy or indicating the presence of DR.

Numerous studies have focused extensively on image pre-processing to extract essential lesions. Local binary pattern (LBP) [56, 57, 62, 84], local ternary pattern (LTP) [62], histogram of oriented gradients (HOG) [62, 85], and scale-invariant feature transform (SIFT) [62, 86] have been widely employed as feature extractor (FE). Those preprocessing techniques positively impacted the performance of the SVM model employed in the study [62], achieving better results on the IDRiD dataset, as shown in Table 5. On the other hand, in the preprocessing stage, the optical disc is often considered as noise that could affect the performance of the model [51]. However, studies have shown that the presence of the optic disc does not significantly impact model performance, as compared to the other studies shown in Table 5.

The CNN model used in these studies [14, 91] required a lot of computational resources and training time. However, some studies have proposed a computationally efficient model that resembles existing architectures [95]. The model is built with four dense convolutional blocks and utilizes a shortcut or skip connection to the dense block to maintain the gradient during backpropagation [92]. The proposed model [92] has small parameters, making it a lightweight model as shown in Table 5.

Instead of building a CNN model from scratch, pre-trained CNN models are utilized for the binary classification of DR [87–89, 93]. Those pre-trained models employed in the binary classification of DR are EfficientNet [87], ResNet [88], Inception-V3 [89], and DenseNet [93], in which they are not the same in architecture, but use the same basic operation of CNN. The studies [87–89, 93] adopted a pre-trained model with a substantial number of trainable parameters, as shown in Table 5, and training the model on a relatively small dataset raises scientific concerns. Furthermore, some studies compared the performance of the pre-trained model on

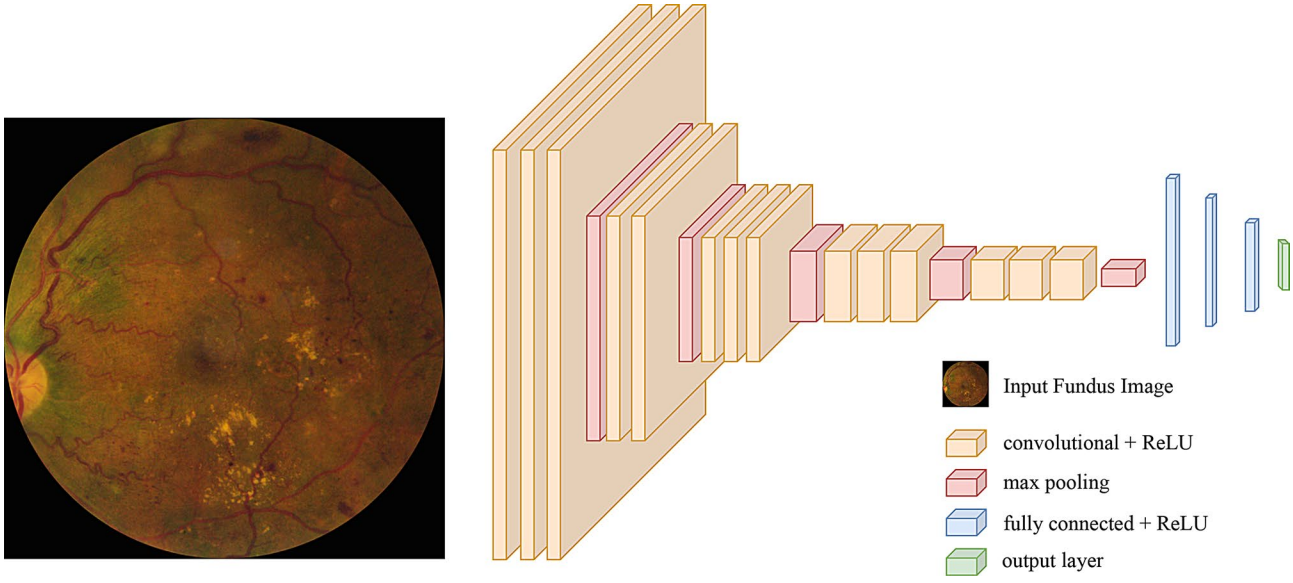


Fig. 10 Commonly used CNN architectures for diabetic retinopathy screening

Table 5 Summary of research on binary classification of DR, including dataset, pre-processing techniques, model employed, and the performance

Author and Year	Dataset	Pre-processing Techniques	Models	Performance
Chetoui, et al., 2020 [87]	EyePACS and APTOS 2019	Blurring, Resize	EfficientNet-B7	Sensitivity: 98.1% Specificity: 93.7% AUC: 0.984
Ghan, et al., 2020 [51]	IDRiD	Standardization, Green Channel Extraction, AHE, and Region of Interest Extractions	R-CNN	Accuracy: 93% F1-score: 0.92
Bibi, et al., 2020 [62]	STARE, DRIVE, E-OPHA	Resizing, Grayscale conversion, contrast enhancement	SVM	Accuracy: 96.6% Sensitivity: 96.4% Specificity: 96.9%
Hongyang, et al., 2020 [88]	MESSIDOR and private dataset	CLAHE, Resize, Augmentation	Redesigned ResNet50	Sensitivity: 93.9% Specificity: 94.4%
Sílvia Rêgo, et al., 2021 [89]	EyePACS	Resize, Augmentation	Inception-V3	Sensitivity: 81% Specificity: 97%
Gurcan et al., 2021 [90]	MESSIDOR-2	Resizing and Normalization	Inception-V3 (23.9 M) with XGBoost	Accuracy: 91.40% AUC: 93.53%
BK Anoop, et al., 2022 [91]	APTOS 2019	Intensity conversion, Denoising, Contrast enhancement, Resizing, and Augmentation	CNN	Sensitivity: 86% Specificity: 96% Accuracy: 94.6%
Bala, et al., 2022 [92]	APTOS 2019	Resizing and Normalization	CNN	Accuracy: 97.54%Precision: 97.55%F1-Score: 0.9747
Nandakumar, et al., 2022 [93]	APTOS 2019	Ben Graham Techniques	DenseNet-121	Accuracy: 96% Precision: 93.51% Recall: 98% F1-Score: 0.98
Begriche, et al., 2023 [78]	APTOS 2019	Resizing and Gaussian Filtering	fine-tuned XCeption	Accuracy: 99.8%
Uzair, et al., 2023 [94]	EyePACS	Resizing, Augmentation, Median filter, Image Sharpening	CNN with SVM	Accuracy: 98.85% Sensitivity: 98.85% Specificity: 99.71%

retinal fundus image classification [78]. DenseNet121, MobileNetV2, and XCeption models are compared, and a fine-tuned XCeption outperformed DenseNet121 and MobileNet, achieving 99.8% accuracy [78] and performs better than other binary DR studies on APTOS 2019 dataset as shown in Table 5.

In binary classification, existing research consistently highlights strong model performance. This indicates a high separability between the two classes, affirming the models' ability to identify and distinguish the normal class from the DR-positive classes [51, 62, 78, 87–89, 91–94].

Diabetic retinopathy grading

The objective of DR grading is to categorize retinal fundus images into different grades. Although most of the studies used five grades, some used three or four grades, and even seven grades. The literature on DR grading is discussed in the following subsection.

Custom CNN models

Custom CNN models are built entirely from scratch or adapted from pre-trained models with significant modifications. Studies [68, 96–98] employed custom-designed CNNs.

The integration of a custom-designed CNN model with techniques such as active DL [99] and siamese networks [100] presents a robust solution for image classification tasks, particularly in scenarios with limited data availability. Active DL, as demonstrated in the context of DR classification [98], enables the algorithm to actively select informative patches from retinal fundus image

grids, optimizing model performance. Furthermore, to address dataset scarcity, the integration of siamese networks in a custom-designed CNN, along with the hierarchical clustering of image patches for local and global feature extraction [101], further enhances the model performance. In particular, the CNN model integrated with active DL achieved better performance [98], showcasing the efficacy of the combined approach in overcoming data limitations and improving classification outcomes with better performance on the EyePACS dataset as shown in Table 6.

Attention mechanisms are used for enhancing the effectiveness of DR classification, especially considering the tiny nature of DR lesions. Using residual attention modules and the spatial pyramid system helps extract relevant features from retinal fundus images [103]. Furthermore, employing a cascaded network that incorporates spatial and loss-based multibranch attention enables the model to capture the information of the lesion [97]. To extract even more informative features, techniques such as dense block, asymmetric convolution branches, channel attention, and auxiliary classifiers are utilized [79]. In contrast, a custom CNN model inspired by attention-based bidirectional long- and short-term memory (ABiLSTM) and the Komodo MLIPIR algorithm (KMA) showcases computational efficiency, minimizing processing time while achieving better performance [105, 107] as illustrated in Table 6. The integration of attention mechanisms and model architectures [105] underscores significant contributions in DR classification, which

Table 6 Summary of research on DR grading using custom-designed CNN, including dataset, pre-processing techniques, model employed, and the performance

Author and Year	Dataset	Pre-processing Techniques	Models	Performance
Hossen et al., 2020 [102]	APTOS 2019	Ben Graham, Resizing, and Augmentation	Deep CNN	Accuracy: 94.49% Precision: 95% Recall: 92% F1 Score: 0.93
Qureshi, et al., 2021 [98]	EyePACS	Normalization and Augmentation	Active DL CNN	Accuracy: 98% Sensitivity: 92.20% F1-score: 0.93
Gayathri, et al., 2021 [96]	IDRiD, MESSIDOR	Resizing	Multipath-CNN and ML Classifier	Accuracy IDRiD: 99.03% and MESSIDOR: 99.75%
Fan, et al., 2021 [103]	APTOS 2019	Cropping and Resizing	Residual Attention Module Powered CNN	Accuracy: 85.32% AUC: 0.8532
Sikder et al., 2021 [104]	APTOS 2019	Resizing, Augmentation, GLCM, Histogram feature extraction	XGBoost	Accuracy: 94.20% Precision: 94.34% Recall: 92.68%
Ashwini, et al., 2022 [67]	IDRiD, DDR, and EyePACS	Resizing, DWT	CNN	Accuracy IDRiD: 90.07%, DDR: 96.20% and EyePACS: 93.53
Cao, et al., 2022 [79]	EyePACS	Optic Disc Removal, Gaussian Filtering, Weighted Image Fusion, and Resizing	Densely Connected Asymmetric CNN	Accuracy: 88.8% Specificity: 97.1% Sensitivity: 88.7%
Yue, et al., 2023 [97]	APTOS 2019 and EyePACS	Resizing, Normalization, and Augmentation	Attention Driven Cascaded Network	Accuracy: 72.44%
Nahiduzzaman, et al., 2023 [68]	APTOS 2019	CLAHE, Normalizations and Resizing	Parallel CNN and ELM	Accuracy: 97.27% Recall: 95% Precision: 96% AUC: 0.9889
Abiram, et al., 2023 [105]	MESSIDOR	Resizing and Normalization	KMA-optimized ABiLSTM	Accuracy: 99.36% Precision: 99% Specificity: 98% Recall: 98%
Mercaldo, et al., 2023 [106]	EyePACS and APTOS 2019	Guassian Filtering, Cropping, and Augmentation	Robust and Explainablel CNN	Accuracy: 91%

improve both performance and computational efficiency as indicated in Table 6.

Global channel attention(GCA)-EfficientNet (G-ENet), a lightweight model for DR classification that integrates a GCA mechanism into EfficientNetB0 [108]. The GCA mechanism combines Global Max Pooling with one-dimensional convolution and a fully connected layer to improve spatial feature embedding and global cross-channel interaction [108]. Replacing MBConv modules with GBConv reduces computational complexity while maintaining high accuracy, with G-ENet outperforming models like ResNet-50 and DenseNet-121 [108]. On the other hand, SE-MIDNet, a DL framework for DR classification that integrates an improved Inception module, dense connections, and a Squeeze-and-Excitation (SE) module [109]. The improved Inception module enhances multi-scale feature extraction by utilizing smaller convolution kernels, which reduces computational complexity while maintaining effective feature representation [109]. Dense connections facilitate feature reuse and mitigate gradient vanishing, enhancing the model's efficiency [109]. The SE module dynamically weights feature channels, allowing the network to prioritize significant features [109]. The end-to-end approach minimizes preprocessing complexity, making SE-MIDNet computationally efficient and suitable for real-world clinical applications in automated diabetic retinopathy classification [109].

Siamese Convolutional Neural Network (SCNN) that analyzes pairs of eye fundus images to measure their similarity has been employed [110]. This architecture, which shares weights across two identical networks, enables effective few-shot learning, allowing the model to perform well with limited labeled data [110]. Furthermore, SCNN is robust against data imbalances, focusing on anomaly detection to identify early signs of DR [110].

In the classification of DR, a hybrid approach that combines a custom-designed CNN for feature extraction and ML classifiers for classification has shown notable efficacy [68, 96]. The extreme learning machine (ELM) is used after feature extraction using parallel convolutional layers [68]. As shown in the Table 6, parallel CNN with ELM shows better performance on the APTOS 2019 dataset. Although multi-path CNN is used for extracting both local and global features, followed by classification using SVM, random forest, and J48 algorithms [96]. Remarkably, the J48 algorithm outperforms SVM and random forests. This integration of ML classifiers on parallel and multi-path CNN architectures not only yields promising performance but also offers reduced computational complexity, highlighting its potential for robust and efficient DR classification [68, 96].

Furthermore, graph neural networks (GNN) are used to capture complex relationships within deep features, thereby learning robust representations of graph-structured data [111]. These networks have been seamlessly integrated into CNN architectures to capture the relationships among features extracted from retinal fundus images through CNNs [86]. However, utilizing GNNs requires significant computational resources and extensive datasets to effectively learn the dependencies, particularly evident in the context of analyzing retinal fundus images [86].

Although a custom-designed CNN has been evaluated with different wavelet transformation techniques, such as Haar, Daubechies, and Symlet wavelet transformations, across seven grades of DR [112], the study demonstrates that Symlet wavelet transformation pooling significantly enhances DR classification. On the other hand, a deep CNN model that resembles the DenseNet-121 model, includes three dense blocks, transition layers, global average pooling, and dropout layers to enhance feature reuse, reduce parameters, and prevent overfitting, has been evaluated and shown a promising result [102]. Despite the computational complexity, the proposed model performs better on the EyePACS dataset [102] as shown in Table 6.

On the other hand, study [104] relied on handcrafted features rather than DL-based automatic feature extraction. Gray Level Co-occurrence Matrix (GLCM) was used to capture texture patterns in retinal images, while histogram-based features provided intensity distribution information [104]. By feeding these extracted features into the XGBoost classifier, the study aimed to achieve efficient and interpretable classification of DR severity. The results indicated that traditional feature engineering, when combined with a powerful ensemble learning algorithm like XGBoost, could yield competitive performance compared to deep learning models, especially in scenarios with limited training data or computational resources [104].

In conclusion, the custom CNN models for DR grading [98, 101] showcase diverse methodologies and approaches for boosting the performance of the model and building a lighter model. Although some models achieve remarkable accuracy rates, challenges persist in terms of computational demand and model complexity.

Transfer learning

In ML, transfer learning is a technique whereby knowledge gained from one task is applied to improve performance on a related task. Similarly, a pre-trained CNN model is used in several DR grading studies [56, 70, 72, 113–129].

As the lesion of DR is highly inseparable, studies [63, 130, 131] emphasize enhancing the techniques used to improve the quality of images. While resizing the retinal fundus image, important features or lesions might vanish.

To ensure that some important features remain intact, downscaling techniques are used to selectively retain essential features, preventing their potential loss [131]. Nevertheless, it is important to note that not all information is preserved through the downscaling process. On the other hand, studies [63, 130] re-categorized inseparable classes of DR. This not only deals with the complexities of the data but also makes things less complicated for the model. The thoughtful approach to both image preparation and model design shows a careful way to handle the challenges in DR grading [63, 130].

The VGG model was used for the classification of DR in various studies [70, 113–115, 132, 133] as it is a powerful classification tool. The hierarchical VGG model used to identify DR and then grade its severity [114]. The approach reduces confusion for the model. A stacked spatial pyramid pooling layer and a network-in-network(NiN) layer to make the model highly nonlinear and scale-invariant deep model [115]. However, VGG demands lots of computing power due to its parameters, as shown in Table 7. To handle computational resource issues, a genetic algorithm (GA) [135] is introduced to the VGG model [132]. Using GA with the VGG model boosts the performance of the model and reduces the computational resources required for training [132].

Pretrained models such as VGG16 and ResNet50 have been evaluated on the IDRiD dataset by incorporating dropout layers in the fully connected layers to mitigate overfitting [134]. While these models demonstrated promising performance, their deployment is constrained by the substantial number of parameters as indicated in the Table 7, making them less practical for resource-limited devices.

SqueezeNet architecture [136] is designed for efficient inference on devices with limited resources, such as mobile devices or embedded systems. SquuezeNet is used for feature extraction enhanced with class attention [137]. As SqueezeNet is sensitive to hyperparameterization, the

Table 7 Summary of research on DR grading using transfer learning, including dataset, pre-processing techniques, model employed, and the performance

Author and Year	Dataset	Pre-processing Techniques	Models	Performance
Vaishnavi, et al., 2020 [52]	MESSIDOR-2 and STARE	Green channel extraction, CLAHE, CLAHE based Segmentation	AlexNet	Accuracy: 95.86% Sensitivity: 92%
Doshi, et al., 2020 [131]	IDRiD	Cropping and Downscaling	InceptionV3	Accuracy: 85.2% Sensitivity: 83.4%
Khaled, et al., 2020 [114]	EyePACS	Normalization and Resizing	VGG-16	Accuracy: 65.5%
Yaqoob, et al., 2020 [116]	MESSIDOR-2 and EyePACS	Resizing, Histogram Gradient and Canny	ResNet-50	Accuracy MESSIDOR2: 97.01% and EyePACS: 99.9%
Lee, et al., 2021 [134]	IDRiD	Resizing, CLAHE, Augmentation	VGG16 and ResNet50	Accuracy for ResNet50: 97.21% Accuracy for VGG16: 94.03%
Bhardwaj, et al., 2021 [69]	MESSIDOR-2	Resizing, Median Filter, CLAHE, Normalization, and Augmentation	Inception-V3	Accuracy: 93.33%
Rajkumar, et al., 2021 [118]	EyePACS	Resizing and Image thresholding	ResNet-50	Accuracy: 99.15% Precision: 98.94%
Martinez, et al., 2021 [120]	MESSIDOR-2	Normalization, Resizing, and Cropping	ResNet-18 and ResNet-50	Accuracy ResNet-18: 91.2% and ResNet-50: 93.6%
Butt, et al., 2021 [130]	APTOS 2019	Resizing and Normalization	ResNet-18 and GoogleNet	Accuracy: 89.29%
Bilal, et al., 2022 [55]	MESSIDOR-2, and EyePACS	Resizing, Green Channel Extraction, and Top-Bottom Hat Transformation	Inception-V3	Accuracy: 97.93% Precision: 97.44% F1-Score: 0.97
Jabbar, et al., 2022 [70]	EyePACS	Resizing, CLAHE, Weighted Gaussian Blurring	VGGNet	Accuracy: 96.6% AUC: 0.97
Kasim, et al., 2023 [71]	MESSIDOR-2, APTOS 2019, and DDR	Resizing, Median Filtering, and CLAHE	MobileNet	Accuracy: 96.06%
Athira, et al., 2023 [63]	APTOS 2019	Resizing, Gaussian Blurring, and Circular Cropping	ResNet50	Accuracy: 94.7%
Kommaraju, et al., 2023 [72]	DDR	Median Subtraction, Gamma Correction, CLAHE, and Augmentation	DRCNNRB	Recall: 94% Precision: 95% F1-Score: 0.94

coyote optimization algorithm [138] is employed in the research conducted in DR grading [137]. Furthermore, deep ELM is used to classify the extracted features on SqueezeNet [137]. This approach is noteworthy, considering the lightweight nature of SqueezeNet [136] and the strategic integration of the classical ML classifier, which enhances overall performance and reduces computational complexity.

ResNet uses shortcuts that skip one or more layers, allowing the network to learn residual functions [139]. The ResNet architecture was used by many studies [56, 72, 116–118, 120–125, 140] to boost performance as shown in Table 7. Furthermore, to improve the depth of propagation of the gradient and contribute to a more effective training process, the Swish activation function is integrated in ResNet18 [140]. Three residual blocks and one convolutional block are utilized, trimming off the remaining blocks [72] to reduce the number of trainable parameters. This approach aims to simplify the architecture, reducing computational complexity, while still maintaining the effectiveness of the model [72]. A fine-tuned ResNet50 is compared to Inception-V3 by optimizing the last layer of the model, surpassing Inception-V3 [116], as indicated in Table 7. Although ResNet is used for feature extraction and the classical ML classifier for classification [117]. This could be a good approach to utilize the strength of the classical ML classifier for classification and the strength of ResNet for feature extraction. Gradient boosting [117], random forest [141], and SVM [80, 123, 133] are among the ML classifiers that are used so often. The proposed model [117, 124] requires huge computational resources and inference time, as it has very high trainable parameters. However, the study [116] demonstrated a high performance on the MESSIDOR-2 dataset, as shown in Table 7.

MobileNet is a family of lightweight CNN architectures optimized for efficient deployment on mobile and embedded devices [141]. An ensemble classifier, namely a random forest to classify the features extracted via MobileNet [141]. On the other hand, densely connected networks incorporate distinctive connectivity in deeper layers known as dense connections [95]. The advantages of these networks over ResNets [139] and InceptionNets [142] are particularly in terms of enhanced supervision, reduced complexity, parametric efficiency, lower computational power requirements, and higher accuracy [95]. Densely connected networks optimized using hyperparameterization for DR grading [126–128]. Although MobileNet [141] would be better than DenseNet [126–128] in a resource-constrained environment like a mobile device.

The InceptionV3 architecture is a deep CNN comprising multiple convolutional layers organized into modules called inception modules [142]. In DR classification, InceptionV3 is utilized to capture lesions or features on different scales using inception modules [55, 69, 131]. Multichannel InceptionV3 is performed on four patches extracted from input retinal fundus images to extract features and then concatenate them [131]. This approach aims to keep gradients going deeper without vanishing [131]. A quadratic-based modification is also introduced for the pre-processing and patching of input images into four to enable the InceptionV3 model to extract features from tiny lesions in DR grading [69]. InceptionResNetV2, a hybrid deep neural network architecture that combines elements from both the Inception and ResNet architectures, is also utilized in DR grading [143]. Additionally, a multi-scaling layer is added on top of the Inception-ResNet structure [143].

Furthermore, transfer learning models and data enhancement techniques, such as CLAHE and Gaussian filtering, for DR classification using various convolutional neural networks are evaluated [144]. EfficientNet-B0 demonstrated the best performance, benefiting from its minimal parameters and reduced computational complexity achieved by scaling depth, width, and resolution, as indicated in Table 7. Compared to ResNet-50, DenseNet-201, and SeResNet-50, EfficientNet-B0 offers a balance between efficiency and accuracy, making it suitable for resource-constrained devices like mobile and edge computing [144]. The findings highlight the potential of deep learning to enhance early detection and intervention in DR [144].

Ensemble techniques

Multiple ML models are used in ensemble approaches to create an ideal predictive model for improved prediction performance [74, 145–148]. Ensemble methods encompass bagging, boosting, and stacking, where bagging aggregates multiple models' results for a generalized outcome, boosting sequentially improves model stability by addressing errors, and stacking integrates predictions from diverse models to construct a new model [148].

VGG16 and VGG19 stacked to enhance the overall performance of the model in grading retinal fundus image [148]. However, the performance of the model is suboptimal. Although the VGG model is ensembled with XCeption using multiple dual attention [149]. The multiple dual attention, the first attention emphasizes the lesion-specific regions, while the second attention facilitates the model in learning correlations among the spatial features descriptor [149].

DenseNet is ensembled with EyeNet for extracting features from the retinal fundus images [145]. As both DenseNet and EyeNet have a skip connection, the gradient will flow efficiently through the network layers, enabling better information propagation and enhancing the model's ability to capture complex lesion patterns and

features from the retinal images [145] as shown in Table 8. On the other hand, the DenseNet ensemble with other heavy models like Xception, InceptionV3, and ResNet50 [147]. Additionally, deformable registration was utilized on the retinal images that reserve all the space and illuminate the effect of the background images of the dataset [147].

Furthermore, five pre-trained models have been ensembled to boost the performance of the model by extracting very sophisticated lesion patterns [146]. The adopted models are ResNet50, InceptionV3, XCeption, DenseNet121, and DenseNet169, which are very heavy and require high computational resources [146]. Despite the model's computational demand, they train the model by extracting and training the model on a single channel of retinal fundus images [146]. This approach aims to enhance the visibility of the lesions by concentrating on each channel [146]. The study [56] took a different approach, by using a feed-forward network with hybrid features from GoogleNet [142], Fuzzy color histogram (FCH) [150], GLCM [84], and LBP, followed by ResNet-18. They also utilized feature dimension reduction, which is Principal Component Analysis (PCA) [56]. Despite the computational complexity of the models, the ensemble of ResNet-18 and GoogleNet [56] demonstrates superior performance on the APTOS 2019 dataset, as shown in Table 8.

Using the ensemble CNN model for feature extractions and classical ML classifiers for classification has been a good approach [74]. DenseNet, GoogleNet, and ResNet combined to extract features and deep ELM for classification of extracted features [74]. However, significant computational resources are essential for both the training and inference phases of these models [74] and indicated in Table 8.

In evaluating these ensemble approaches, it becomes crucial to consider the appropriateness of the models that are combined, along with the computational resources and training time required. While combining diverse models can potentially yield good results, understanding their contribution and the associated resource implications is essential for practical implementations [145, 146].

Capsule network

The advent of Capsule Network [151] enhances the encapsulation of entity relationships, including scale, location, pose, and orientation, a capability lacking in traditional CNN models. A convolution layer, primary capsule layer, and class capsule layer with dynamic routing [152] have been utilized within the capsule network [73]. Although the VGG network has been combined with the capsule network [153]. Capsule networks with dynamic routing have a better performance [152]. Despite Capsule Network's strengths [152, 153], it is resource-intensive and demands a huge training time. As shown in the Table 9, the capsuleNet has huge trainable parameters, which makes the model computationally resource-intensive.

Table 8 Summary of research on DR screening using ensemble techniques, including dataset, pre-processing techniques, model employed, and the performance

Author and Year	Dataset	Pre-processing Techniques	Models	Performance
Abdelmaksoud, et al., 2020 [145]	APTOS 2019	Cropping, Resizing, and Normalization	EyeNet and DenseNet121	Accuracy: 91.6% Dice: 92.45% Kappa: 0.883
Harihanth, et al., 2020 [146]	EyePACS	Resizing, Separate RGB channels and Stacking, and Augmentation	ResNet50, XCeption, InceptionV3, DenseNet121, and DenseNet169	Accuracy: 81.85%
Novitasari, et al., 2022 [74]	DRIVE and MESSIDOR	Cropping, CLAHE, Resizing, and Augmentation	Deep ELM	Accuracy: 98.20%
Oulhadj, et al., 2022 [147]	APTOS 2019	Resizing and Deformable Registration using B-Spline Approach	DenseNet121, XCeption, InceptionV3, and ResNet50	Accuracy: 85% Precision: 80% Recall: 70% F1-Score: 0.73
Alshahrani, et al., 2023 [56]	APTOS 2019	Green Channel Extraction, Resizing, Laplacian filtering, FCH, GLCM, and LBP	ResNet-18 and GoogleNet	Accuracy: 99.7% Precision: 99.6% Sensitivity: 99.6% Specificity: 100% AUC: 0.9986
Kale, et al., 2023 [148]	APTOS 2019	Resizing and Augmentation	VGG16 and VGG19	Accuracy: 87.31% Precision: 80% Recall: 87% F1-Score: 0.89
Bodapati, et al., 2024 [149]	APTOS 2019	Resizing	VGG16 and XCeption	Accuracy: 86.22% Kappa: 89.65% AUC: 0.9647

Table 9 Summary of research on DR screening using capsule networks, including dataset, pre-processing techniques, model employed, and the performance

Author and Year	Dataset	Pre-processing Techniques	Models (parameter)	Performance
Kumar, et al., 2021 [153]	APTOS 2019	Resizing and Augmentation	Capsule Network and VGG16	Accuracy: 75.8%
Kalyani, et al., 2023 [73]	MESSIDOR	Resizing, CLAHE, and Gaussian Filter	Capsule Network	Accuracy: 97.98%

Table 10 Summary of research on DR lesion localization, including dataset, pre-processing techniques, model employed, and the performance

Author and Year	Dataset	Pre-processing Techniques	Models	Performance
Wan, et al., 2021 [155]	IDRiD and E-Ophtha	Resizing, Removing Redundant Black Edge	Attention Decoder Network	Accuracy: 99.97% F1-Score: 0.9089
Nazir, et al., 2021 [156]	APTOS 2019 and IDRiD	–	DenseNet-100 and CenterNet	Accuracy APTOS2019: 97.93% and IDRiD:98.1%
Saranya, et al., 2023 [54]	MESSIDOR and E-Ophtha	Resizing, Green Channel Extraction, CLAHE, Guassian Filtering, and Gamma Corrections	CNN	Accuracy MESSIDOR:97% and E-Ophtha:6.32%
Vatsal Agrawal et al., 2024 [157]	IDRiD, HEDDB, AIIMSDB	Histrogram Equalization, Normalization, ISO noise Removal	YOLO-V5	Dice coefficient: 0.84

Diabetic retinopathy localization

DR localization aims to precisely classify, localize, and map retinal lesions region, which is crucial for diagnosing DR.

Detecting the precise location of a lesion and assisting ophthalmologists can be achieved by first determining the color of the lesion and then localizing each specific color [154]. On the other hand, the CNN model was designed with encoder, attention, and decoder modules, and incorporating attention helps to yield superior results [155]. Studies [154, 155] encourage lightweight models and explore more complicated architectures that can maximize the model performance, like attention mechanisms as shown in Table 10.

YOLOv3 is integrated with a feature pyramid network for the prediction of bounding boxes that are manually annotated during retinal fundus image pre-processing [158]. YOLOv3 [159] is a DL object detection model that works by dividing the input image into a grid and then predicting bounding boxes and class probabilities for objects within each grid cell. These bounding boxes help the model focus on specific features within the box. The model's efficiency results in relatively good inference time [158]. Although a custom DL CenterNet model is used for DR grading. This model generates annotations for precise lesion localization and uses DenseNet-100 for feature extraction, followed by CenterNet for detection [156]. However, those models are heavy and require huge computational resources [156, 158] as indicated in Table 10.

Furthermore, the Multi-Level EfficientNet-B0 architecture is utilized, in which it captures fine-grained details by integrating feature maps from various layers, improving the detection of subtle signs of DR [160]. The study also employs ensemble learning to average predictions from multiple models, enhancing sensitivity and specificity [160] as shown in Table 10. A sophisticated color normalization technique standardizes image inputs, and the use of Test-Time Augmentation (TTA) during inference allows for improved performance by generating and averaging predictions from multiple augmented image versions [160].

NetraDeep employs a hybrid approach that integrates DL and rule-based image processing techniques for the precise detection of hard exudates in retinal images [157]. It utilizes a YOLO-V5 model for detecting retinal features, complemented by a dynamic region-based modified OTSU algorithm for exudate segmentation [157]. This approach effectively addresses the challenges posed by low-quality images, making it suitable for clinical applications [157].

Diabetic retinopathy segmentation

Segmentation techniques are essential for identifying the region of interest in DR screening. The regions of interest may include the optic disc, fovea, and blood vessels, lesions such as microaneurysms, hemorrhages, soft exudates, and hard exudates in the retinal fundus image. Manual segmentation techniques are laborious, resource-intensive, and time-consuming. However, segmentation can be easily carried out using ML models [57–60, 64, 65, 75–77, 161–166].

Blood vessel segmentation

One of the techniques employed in DR segmentation is abnormal blood vessel segmentation, as abnormalities in blood vessels might be caused by DR. Several studies used blood vessel segmentation techniques to identify the region of interest for blood vessel abnormalities.

Retinal vascular tortuosity is an excessive bending and twisting of the blood vessels in the retina that is associated with numerous health conditions [167]. The paper [167] used UNet-based automated assessment of retinal vascular tortuosity using color fundus images. The model extracts the arteriovenous tree, optic disc, and fovea using specialized neural networks and calculates tortuosity scores by incorporating anatomical factors such as vessel caliber, artery-vein distinction, and distances to key structures [167].

The study [168] focuses on diagnosing hypertensive retinopathy using CCN with an encoder-decoder architecture for artery-vein classification. The model employs a multi-loss function to optimize both pixel-wise and segment-wise classification, enhancing accuracy [168]. On the other hand, study [169] introduces an improved method for extracting vessel segments from retinal fundus images, addressing issues in traditional skeletonization techniques. Using morphological operations and optimized skeletonization, the method enhances vessel continuity, preventing unnecessary segment breaks [169].

Furthermore, study [170] aims to develop an automated system for detecting and grading hypertensive retinopathy using retinal vessel morphometry, specifically arteriovenous ratio (AVR) and vessel tortuosity severity. The proposed model integrates a CNN for artery-vein classification, an iterative AVR calculation method, and machine learning classifiers (Decision Tree and Rotation Forest) for tortuosity grading [170].

Classical ML models are used with a feature extractor, as mentioned in the studies [53, 64]. Average filtering and Otsu thresholding [171] are employed to obtain the binary form of the image, and overlaying techniques are used on the complete image and original RGB images. Additionally, they employed a coherence-enhancing energy-based regularization approach on the green channel of the retinal fundus image to segment blood vessels [53]. They also utilized Canny anisotropic diffusion [172], morphological transformations [173] for optic disc segmentation, and the Harris hawk optimization algorithm [174] for feature optimization.

The UNet model, characterized by its U-shaped design and utilization of encoding and decoding paths with skip connections, has demonstrated considerable effectiveness across segmentation tasks in image processing [175]. UNet was used with significant modifications to make the model lighter and get a better-performing segmentation model [161] that is robust enough, as shown in Table 11. The convolutional block in UNet was reduced from four to three in both the encoder and decoder layers [161]. On the other hand, multipath recurrent UNet is used to enhance discrimination in retinal fundus

Table 11 Summary of research on blood vessel segmentation, including dataset, pre-processing techniques, model employed, and the performance

Author and Year	Dataset	Pre-processing Techniques	Models	Performance
Jiang, et al., 2020 [162]	Drishti-GS1 and DRIVE	Resizing, and Augmentation	Recurrent UNet	Accuracy Drishti-GS1: 99.50%, and DRIVE: 96.42%
Toptacs, et al., 2020 [58]	DRIVE	Green Channel Extraction, Edge, Morphological, Statistical, Hessian, and Gradient-Based Feature Extraction	ANN	Accuracy: 95.63%, Jaccard: 0.6148, and Dice: 0.76
Tamim, et al., 2020 [59]	DRIVE, STARE, and CHASE DB1	Green Channel Extraction, Feature Extraction	MLP	Accuracy DRIVE: 96.07%, STARE: 96.07%, CHASE DB1: 95.77%
Gegundez, et al., 2021 [161]	DRIVE, STARE, and CHASE DB1	_	UNet	Accuracy DRIVE: 95.63%, STARE: 96.35%, CHASE DB1: 95.61%
Tchinda, et al., 2021 [60]	DRIVE, STARE, and CHASE DB1	Green Channel Extraction, Median Filter, Robert Filter	MLP	Accuracy DRIVE: 94.80%, STARE: 95.48%, CHASE DB1: 94.52%
Badawi, et al., 2022 [170]	RVM (504 images)	Green channel extraction, CLAHE, morphological operations, B-COSFIRE filter, OD detection	CNN, Decision Tree (J48), Rotation Forest	Accuracy: 96.82% (HR severity classification), 92.66% (tortuosity classification)
Gargari, et al., 2022 [57]	DRIVE and MESSIDOR	CLAHE, Gaussian Filter, and Gabor Filter	UNet++	Accuracy DRIVE: 98%, and MESSIDOR: 98.6%
Sidhu, et al., 2023 [64]	STARE and PGIMER dataset	Grayscale Conversion, Resizing, Normalization	SVM	Accuracy STARE: 96.10%, and PGIMER: 96.82%
Badawi et al., 2023 [168]	AV-Classification dataset	Optic Disc Localization, ROI Segmentation, Vessel Segmentation, Artery-Vein ClassificationSkeletonization and Vessel Segment Optimization	CNN with encoder and decoder	Pixel-wise accuracy: 97%
Hervella et al., 2024 [167]	hr DRIVE	Resizing	UNet	AUC: 0.95

images and address pathological region issues [162]. The approach combines convolutional and recurrent neural networks, leveraging recurrent networks for temporal feature multipathing [162]. In contrast, the dense residual path UNet is trained on the retinal fundus image of patch 64 × 64 [176]. Utilizing dense residual UNet enhances feature extractions, improves the model performance, and reduces computational complexity [176].

UNet++ is an extension of UNet, offering significant improvements in information preservation and reduction of information loss [177]. In a study [57], UNet++ was used on the green channel of retinal fundus images filtered by the Gabor filter. Furthermore, the segmented image underwent HOG [57, 57] and LBP analysis to form a binary format. A one-dimensional convolutional layer was then utilized for disease diagnosis [57]. The proposed model is lightweight and demands fewer computational resources [57], as indicated in the Table 11.

The study proposed the use of a multilayer perceptron (MLP) with twenty-four input units in the input layer, three hidden layers, and one output layer in order to reduce time complexity [59]. In contrast, a feed-forward network known as MLP is utilized, featuring an additional connection from the input to each layer and from each layer to all subsequent layers [60]. The MLP model proposed in the study [59] exhibited superior performance compared to the model presented in the study [60] and shown in Table 11. Additionally, the time complexity of the model was also enhanced [59].

Lesion-based segmentation

The lesion-based segmentation technique is among the most widely used segmentation mechanisms, as it has various lesions that could be detected in each stage of DR. Those lesions are microaneurysms, hemorrhages, soft exudates, and hard exudates. Studies [75–77] used lesion-based segmentation to identify the type of lesion spotted on the retinal fundus image.

Multi-scale and multi-level feature fusion networks inspired by UNet and incorporating VGG-16 in the encoder enhance the feature learning and utilize transposed convolutional upsampling in the decoder to enhance the decoding capability of the model [178]. Although study [179] integrates advanced DL architectures for DR lesion segmentation and classification, leveraging Xception with DeepLabv3 for segmentation and EfficientNet-b0 with SqueezeNet for feature extraction [179]. The extracted features are optimally selected using the Marine Predators Algorithm (MPA), which enhances classification accuracy by reducing dimensionality and preserving discriminative information [179]. The selected features are then fed into KNN and NN classifiers to categorize DR severity into four grades. This approach is significant because it combines powerful pretrained models with an efficient feature selection mechanism, optimizing computational complexity while improving diagnostic accuracy, crucial for automated diabetic retinopathy screening [180].

The segmentation and classification of DR utilize a Deformable Ladder Bi-attention U-Net (DLBUnet) coupled with a Deep Adaptive Convolutional Neural Network (DACNN) [181]. The DLBUnet performs lesion segmentation through an encoder-decoder structure with a central processing module. In the encoder, deformable convolution replaces standard convolution to capture lesion variations by learning offset locations. The central processing module integrates Ladder Atrous Spatial Pyramid Pooling (LASPP) with variable dilation rates to enhance small lesion features and mitigate gridding effects. The decoder employs a bi-attention mechanism with spatial and channel attention to accurately reconstruct lesion contours and edges, improving the segmentation precision [181]. Another approach for lesion segmentation is the use of DR-UNet and an attention-aware convolutional network based on the ResidualNet architecture, which introduces a skip connection between convolutional blocks to enhance the model's performance [122].

The study used a weakly supervised sensitive heatmap framework to interpret CNN models such as EfficientNetB5, DenseNet201, and ResNet50 for lesion segmentation [182]. The research found that the combination of EfficientNetB5 with the weakly supervised sensitive heatmap framework outperformed the other models [182]. The framework achieved an accuracy of 98.65%, surpassing the 95.4% accuracy achieved by the class activation map, as shown in the Table 12. Using a pre-trained model in conjunction with the weakly supervised sensitive heatmap framework proved to be an effective strategy.

A dual-branch architecture, a relation transformer network specifically designed for multi-lesion segmentation [75]. The global transformer block captures small lesion patterns and preserves detailed information in a deep network [75]. A deeper network with a global transformer block truly offers advantages over shallower models in preserving important information in lesion classification [75]. Although a multi-level segmentation method combines SVM with selective characteristics and GA [135, 184]. This combination signifies a more comprehensive and integrated approach, indicating the need for a holistic strategy in lesion classification [184]. Furthermore, a viewpoint-based weighted kernel fuzzy clustering approach, using a kernel-based hyper-sphere density initialization method that employs kernel distance instead of the Euclidean distance [185], is employed for lesion segmentation as kernel distance is non-linear and more robust. It is imperative to critically assess and compare alternative approaches such as multi-level set

Table 12 Summary of research on lesion-based segmentation, including dataset, pre-processing techniques, model employed, and the performance

Author and Year	Dataset	Pre-processing Techniques	Models	Performance
Sambyal, et al., 2020 [183]	IDRiD and E-Ophtha	Resizing, Pixel Annotation	Modified UNet	Accuracy: 99.9% Dice: 0.998
Huang, et al., 2020 [75]	IDRiD and DDR	Resizing, Augmentation, and CLAHE	Transformer based UNet	AUC on EX: 0.6799, HE: 0.4504, MA: 0.21, SE: 0.34
Al-Mukhtar et al., 2021 [182]	MESSIDOR	Normalization, Global Thresholding, Morphological Transform, Gaussian Blur, and Hough circle transofrm	EfficientNetB5	AUC: 0.98
Xu, et al., 2021 [76]	IDRiD	Cropping, Gaussian Filter, CLAHE, and Normalization	Feature Fusion based UNet	Dice: 0.579
Guo, et al., 2021 [77]	IDRiD, E-Ophtha, and DDR	Resizing, Cropping, and CLAHE	Cascade Attentive RefineNet	Accuracy EX: 98.23%, SE: 97.52%, MA: 91.83% and HE: 95.37%
Yan, et al., 2021 [178]	E-Ophtha and IDRID	Resizing and Augmentation	UNet with VGG16 and Transposed Convolutional UpSampling	AUC MA: 0.4393, HE: 0.6411, SE: 0.7597 and EX: 0.8644
Shaukat et al., 2022 [179]	DIARETDB1, E-Ophtha-EX, IDRiD, and MESSIDOR	Resizing	XCeption, DeepLabV3, EfficientNetB0 and SqueezeNet	Accuracy: E-Ophtha-EX, EX: 96%, IDRID, HM: 86%, HE: 85%, MA: 71%, OD: 86%, SE: 97%
Skouta, et al., 2022 [61]	IDRiD and DiaretDB1	Cropping, Green Channel Extraction, and Resizing	Modified CNN UNet	Sensitivity: 80.49% Specificity: 99.68% Accuracy: 98.68%
Maaliw, et al., 2023 [122]	DIARETDB0, DIARETDB1, FIRE	Resizing, Cropping and Augmentation	DR-UNet and ResNet	Accuracy: 87.10%
Durai et al., 2023 [181]	MESSIDOR-2 and MESSIDOR	Segmentation-based feature extraction using DLBUnet	DLBUnet-DACNN	Accuracy: 98.2%, Recall: 0.987, Kappa: 0.993, Precision: 0.98, F1-score: 0.981

segmentation with SVM and GA [184], kernel fuzzy clustering [185], and dual branch architecture [95].

As the lesion plays a crucial role in the identification of DR, there is a study [54] proposed a custom-designed CNN model to classify lesions. Study [54] used a custom-designed CNN model to detect exudates from retinal fundus images for non-proliferative DR. Detecting exudates poses a significant challenge due to their color similarity to the optic disc, necessitating advanced models for accurate identification [54]. As shown in the Table 12, Modified UNet [183] has shown better performance on the IDRiD dataset, whereas Cascade Attentive RefineNet has a huge trainable parameter.

Additionally, a dual-branch encoder-decoder architecture is introduced to utilize two ResNet-18 networks to extract features from retinal images at different time points, enabling simultaneous feature comparison [186]. Central to this approach is the Local-Global Enhancement (LGE) module, which integrates local and global information to capture subtle lesion changes, while a channel attention mechanism enhances important features and improves robustness against normal vascular variations [186].

Recent advancements in lesion-based segmentation have focused on enhancing feature extraction, improving model architectures, and integrating multi-scale and multi-level feature fusion mechanisms. For instance, studies such as [75–77] have employed lesion-based segmentation to identify and classify lesions in retinal fundus images.

While significant progress has been made, several limitations persist in lesion-based segmentation. One major challenge is the variability in lesion appearance, particularly for microaneurysms, which can be easily confused with noise or other retinal structures. Additionally, the performance of segmentation models is highly dependent on the quality of the dataset and pre-processing techniques. As shown in Table 12, studies using the IDRiD dataset have reported varying performance metrics, with Dice scores ranging from 0.579 [76] to 0.998 [183] for different lesions.

Optic disc and fovea based segmentation

There are studies that have performed a segmentation focusing on an optic disc or fovea [65, 165, 166]. Automation in the segmentation of optic disc and localization of optic disc and fovea centers is crucial in computer-aided screening, although it is very challenging due to diverse artifacts and image variability [187].

Spatial color histogram binarization techniques employed for fovea segmentation [166]. This approach uses both spatial and color information, resulting in a

two-dimensional histogram that encapsulates rich information [166]. The model stands out due to its lightweight nature, requiring minimal computational resources [166]. Additionally, FundusPosNet, which incorporated an encoder-decoder architecture inspired by UNet [165]. Despite the differently designed model, FundusPosNet falls short when it comes to accurately detecting the optic disc and fovea in retinal fundus images exhibiting severe anatomical pathologies [165]. This limitation raises concerns about the model's robustness in handling diverse clinical scenarios [165].

A region proposal network to generate multiple optic disc regions was employed for segmentation [65]. The selection of the proposal with the highest probability forms the basis for identifying the region of interest containing the fovea [65]. Subsequently, a three-level cascade CNN is employed for the precise fovea region. While this approach demonstrates competitive performance, a critical evaluation is necessary [65]. Comparing the performance of models introduced by [65, 165, 166], it becomes evident that the model [166] holds a distinctive advantage specifically in situations where computational resources are a concern or where a balance between accuracy and efficiency is crucial. As shown in the Table 13, the proposed models [65, 165, 166] demonstrate high performance, indicating that the optic disc and fovea can be accurately identified and detected.

Model computational complexity

Model computational complexity plays a critical role in determining the feasibility of deploying DL models, particularly in resource-constrained environments such as mobile devices, embedded systems, or real-time clinical applications. One of the most widely used and practical indicators of computational complexity is the number of trainable parameters within a model.

In this section, we systematically compare the trainable parameters of the DL models reviewed in this systematic review. The total number of parameters directly impacts the memory footprint, training duration, and inference latency of a model. While most of the studies included in this review did not explicitly report other aspects of model complexity, such as memory usage, floating-point operations, or inference time, we focus on model parameters as a consistent and available metric.

Accordingly, we discuss the computational complexity in terms of model parameters for those studies that explicitly reported them, as shown in Table 14. This provides a useful basis for evaluating model scalability and assessing their suitability for deployment in real-world scenarios with limited computational resources.

Current AI trends in diabetic retinopathy screening

Recent advancements in AI, particularly DL, have significantly improved the accuracy of DR diagnosis through automated retinal image analysis. However, challenges remain in terms of model interpretability and the computational resources required for training AI models. As a result, federated learning (FL) and explainable AI (XAI) have become active areas of research in AI-enabled DR screening. XAI enhances the transparency and interpretability of DL models, enabling clinicians to better understand model decisions and trust AI-driven diagnoses. Meanwhile, FL allows collaborative model training across institutions while preserving patient privacy and addressing data-sharing concerns. Together, these approaches are transforming DR screening, improving diagnostic accuracy, and expanding access to high-quality retinal datasets.

Explainable AI

Explainable AI (XAI) encompasses a suite of methods and techniques designed to make the decision-making processes of AI systems transparent, interpretable, and comprehensible to humans. This is particularly critical in addressing the "black-box" nature of many modern AI systems, especially those employing DL [188]. While such systems produce highly accurate results, the underlying processes often remain opaque, limiting their adoption in sensitive domains like healthcare [189].

An ensemble-based approach has been used to enhance the explainability of DL techniques in diagnosing DR [190]. This model integrates weights from multiple CNNs to effectively extract significant features from retinal images. It utilizes Gradient-weighted Class Activation Mapping (Grad-CAM) and Shapley Additive Explanations (SHAP) to provide intuitive visual insights into the model's decision-making process, highlighting the critical regions of interest in retinal scans as shown in Table 15. This dual approach not only improves diagnostic accuracy but also addresses the urgent need for

Table 13 Summary of research on optic disc and fovea-based segmentation, including dataset, pre-processing techniques, model employed, and the performance

Author and Year	Dataset	Pre-processing Techniques	Models	Performance
Huang, et al., 2020 [65]	IDRiD and MESSIDOR	Resizing, AHE, and Data Augmentation	Three Level Cascaded CNN	Accuracy Optic Disc: 100%, Fovea: 99.25
Bhatkalkar, et al., 2021 [165]	IDRiD, G1020, and MESSIDOR	Gaussian Filtering, Resizing, Augmentation, and Heatmap Label Generation	FundusPos Net based UNet	Accuracy on MESSIDOR; Optic Disc: 99%, Fovea: 100%

Table 14 Summary of model computational complexity in terms of parameters reported in Tables 5, 6, 7, 8, 9, 10, 11, 12, and 13

Author	Model	Trainable Parameters
Huang, et al. [65]	Three Level Cascaded CNN	52 M
Sambyal, et al. [183]	Modified UNet	2,056,000
Huang, et al. [75]	Transformer based UNet	2,056,000
Al-Mukhtar et al. [182]	EfficientNetB5	30.6 M
Xu, et al. [76]	Feature Fusion-based UNet	0.97 M
Guo, et al. [77]	Cascade Attentive RefineNet	75 M
Skouta, et al. [61]	Modified CNN UNet	15 M
Gegundez, et al. [161]	UNet	2,056,000
Nazir, et al. [156]	DenseNet-100 and CenterNet	45 M
Kumar, et al. [153]	Capsule Network and VGG16	Capsule Network: 7 M and VGG16: 138.4 M
Kalyani, et al. [73]	Capsule Network	13.8B
Harihanth, et al. [146]	ResNet50, XCeption, InceptionV3, DenseNet121, and DenseNet169	ResNet50:25.6 M, XCeption:22.9 M, InceptionV3:23.9 M, DenseNet121:8.1 M, and DenseNet169: 14.3 M
Novitasari, et al. [74]	Deep ELM	94.8 M
Alshahrani, et al. [56]	ResNet-18 and GoogleNet	ResNet-18: 11 M and GoogleNet: 6.8 M
Kale, et al. [148]	VGG16 and VGG19	VGG16: 138.4 M and VGG19: 143.7 M
Bodapati, et al. [149]	VGG16 and XCeption	VGG16: 138.4 M andXCeption: 22.9 M
Vaishnavi, et al. [52]	AlexNet	62 M
Doshi, et al. [131]	InceptionV3	23.9 M
Khaled, et al. [114]	VGG-16	138.4 M
Yaqoob, et al. [116]	ResNet-50	25.6 M
Lee, et al. [134]	VGG16 and ResNet50	VGG16 138.4 M and ResNet50: 25.6 M
Bhardwaj, et al. [69]	Inception-V3	23 M
Rajkumar, et al. [118]	ResNet-50	25.6 M
Butt, et al. [130]	ResNet-18 and GoogleNet	ResNet-18:11 M and GoogleNet: 6.8 M
Bilal, et al. [55]	Inception-V3	23.9 M
Jabbar, et al. [70]	VGGNet	116.7 M
Kasim, et al. [71]	MobileNet	4.3 M
Athira, et al. [63]	ResNet50	25.6 M
Hossen et al. [102]	Deep CNN	7 M
Qureshi, et al. [98]	Active DL CNN	56,996
Gayathri, et al. [96]	Multipath-CNN and ML Classifier	57,968
Fan, et al. [103]	Residual Attention Module Powered CNN	1 M
Chetoui, et al. [87]	EfficientNet-B7	66.7 M
Hongyang, et al. [88]	Redesigned ResNet50	25.6 M
Sílvia Rêgo, et al. [89]	Inception-V3	23.9 M
Gurcan et al. [90]	Inception-V3 with XGBoost	23.9 M
BK Anoop, et al. [91]	CNN	184.3 M
Nandakumar, et al. [93]	DenseNet-121	8.1 M
Bala, et al. [92]	CNN	1.1 M

interpretability in AI applications within healthcare. By enabling clinicians to understand and trust the AIs reasoning, it facilitates the practical deployment of such systems in ophthalmology, bridging the gap between AI and clinical practice [190].

In DR, XAI techniques significantly enhance the interpretability of DL models used for diagnosis, empowering healthcare professionals to better understand the rationale behind predictions. For instance, these methods enable the extraction of pathological descriptors and the visualization of neuron activation patterns. Such insights establish clear links between specific retinal lesions and model outputs, fostering greater trust among clinicians and improving the overall quality of clinical decision-making. Furthermore, XAI facilitates the generation of diverse synthetic images, contributing to the robust training of diagnostic algorithms and ensuring their applicability across varied patient populations [197].

The XAI-enabled teleophthalmology for DR grading and classification model introduced a unique approach to DR grading and classification through XAI within a teleophthalmology framework [191]. It combines advanced

Table 15 Summary of explainable AI and federated learning on diabetic retinopathy screening, including dataset, pre-processing techniques, model employed, and the performance

Author and Year	Dataset	Pre-processing Techniques	Models	Performance	Category
Shorfuzzaman et al. 2021 [190]	APTOS 2019	Gaussian Blurring, Resizing, Normalization, and Augmentation	CNN Weighted Ensemble with Grad-CAM and SHAP	Accuracy: 96.2%, Sensitivity: 94%, Precision: 97%, AUC: 0.978	XAI
Obayya et al., 2022 [191]	APTOS 2019	Median Filtering and Histogram Equalization	UNet followed by SqueezeNet (1 M), AOA and BGRCU	Accuracy: 96%, Precision: 86.46%, Recall: 83.14%	XAI
Singha et al., 2023 [192]	APTOS 2019	Resizing	Modified DenseNet	Precision: 89%, Recall: 90%, and Accuracy: 94.2%	FL
Chetoui et al., 2023 [193]	APTOS 2019	Resizing	FL-ViT	Accuracy: 95%, Specificity: 95%, Sensitivity: 95% and AUC: 0.95	FL
Singha et al., 2023 [192]	APTOS 2019	Resizing, Standardization, and CLAHE	Enhanced DenseNet	Accuracy: 94.27%, Precision: 89%, and Recall: 90%	FL
Bhulakshmi et al., 2024 [194]	MESSIDOR	Augmentation and Resizing	FedDEO	Accuracy: 96.98%	FL
Raj et al., 2024 [195]	DDR, APTOS 2019, and EyePACS	Resizing	EfficientNet	Accuracy: 93.2%	FL
Bhulakshmi et al., 2024 [196]	IDRiD	Resizing, Normalization and Augmentation	FedDL	Accuracy: 94.66%	FL

techniques such as U-Net for image segmentation, SqueezeNet for feature extraction, and the Archimedes Optimization Algorithm (AOA) [198] and bidirectional gated recurrent convolutional unit (BGRCU) for enhanced detection and classification [191]. This model emphasizes transparency in AI decision-making, enabling clinicians to better understand and trust the results, while also improving access to eye care via remote monitoring. Its effectiveness is validated through rigorous testing on benchmark datasets, demonstrating significant improvements over existing methods [191] and shown in Table 15.

Federated learning

Federated learning is a decentralized approach to training ML models, where sensitive data remains localized at its source, such as hospitals or clinics, instead of being aggregated into a central repository [199]. Models are trained locally on distributed nodes, and only model updates (weights or gradients) are shared and aggregated to build a global model [199].

In DR, FL facilitates collaboration among healthcare institutions without exposing patient data, addressing privacy concerns and regulatory challenges [200]. This enables the development of robust models trained on diverse datasets, improving generalizability and fairness in diagnosis. Studies have shown FLs effectiveness in DR screening, achieving comparable performance to centrally trained models while maintaining data security [194].

The FL approach, combined with ViT, improves the accuracy and efficiency of DR detection across institutions while preserving data privacy [193]. By allowing institutions to collaboratively train a model without sharing sensitive data, this framework enhances privacy and improves classification accuracy by up to 3% [193] as indicated on Table 15. ViT's attention mechanism further boosts performance, and FL-ViT has been shown to outperform local models across diverse datasets while enabling multi-institutional collaboration [193].

Instead of centralizing patient data, FL allows institutions to train local models, leveraging data from multiple sources to improve generalizability across diverse populations and image qualities [195]. This approach is efficient in resource-limited environments [195]. Comparative studies of CNNs like ResNet50, DenseNet201, AlexNet, EfficientNetB7, and VGGNet19 in FL environments highlight their performance [196]. Enhanced DenseNet architectures, incorporating dense blocks to improve feature utilization and address the vanishing gradient problem, further strengthen feature propagation [192]. As shown in Table 15, FL demonstrates promising performance, particularly for resource-constrained environments and ensuring data privacy.

Discussions

Utilizing the search strategies described in Sect. 3.2, we extracted various studies focusing on DR localization, grading, and segmentation that use retinal fundus images. Among the extracted articles and conference papers, we carefully selected seventy eight (78) research papers by using inclusion and exclusion criteria outlined in Table 3. Figure 11 provides summaries of scholarly papers reviewed based on their publication years, while Fig. 12 presents the distribution of articles and

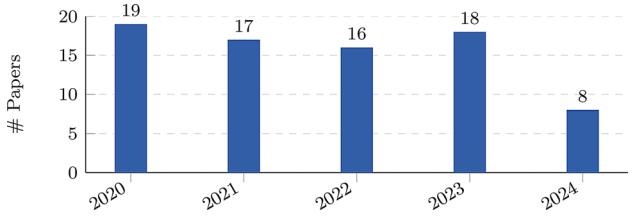


Fig. 11 Yearly distribution of the reviewed research papers

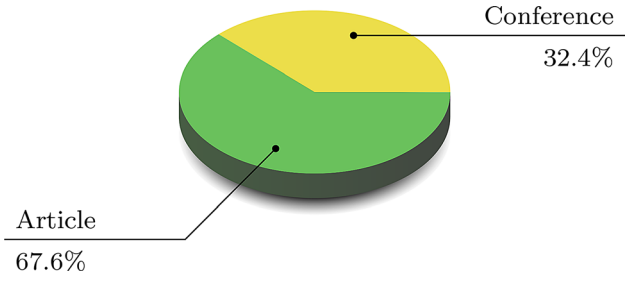


Fig. 12 Distribution of reviewed papers across articles and conferences

conference papers we have reviewed in this systematic review work.

As we discussed in Sect. 4, a commonly used dataset in screening for DR is explained. Figure 13 shows how much the dataset is used in research papers. Depending on the scholarly papers we have reviewed, APTOs, IDRiD, and EyePACS are the top three datasets used by researchers.

In Sects. 7.1, 7.2, 7.4, we discussed the literature about DR binary classification, grading, and segmentation. The model is evaluated on either a single dataset or a mixed dataset consisting of more than one dataset. Assessing the model's performance across multiple datasets helps to know the model's robustness. Figures 14, 15, and 16 shows the top two models identified in binary classification, grading, and segmentation accordingly. We have evaluated the accuracy of the models proposed in each paper, considering factors such as sensitivity, specificity, and overall performance in DR diagnosis. However, our evaluation did not explicitly consider the complexity of the models employed. In DR binary classification models like XCeption [78], custom CNN [92], FE-SVM [62], R-CNN [51], CNN-SVM [94] and Inception-V3 [89] showed good performance on each dataset as highlighted on Fig. 14. Despite their performance, models such as XCeption [78] and Inception-V3 [89] possess a substantial number of parameters, whereas custom CNNs [92] have fewer trainable parameters, necessitating less computational power. In DR grading models such ResNet-18 GoogleNet [56], ResNet-50 [116], active DL CNN [98], multipath-CNN [96], ABiLSTM [105], CNN [67], Inception-V3 [55], and AlexNet [52] showed promising performance on each dataset as shown on Fig. 15, but the models [55, 56, 116] are computationally intensive. Additionally, in DR segmentation models such as ANN [58],

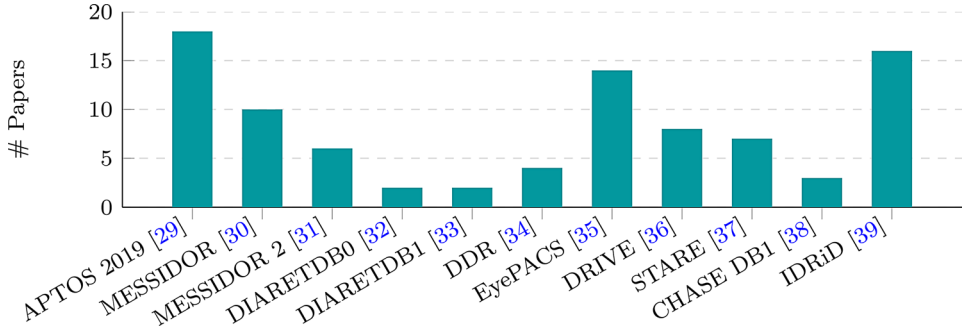


Fig. 13 Distribution of datasets in the systematically reviewed papers

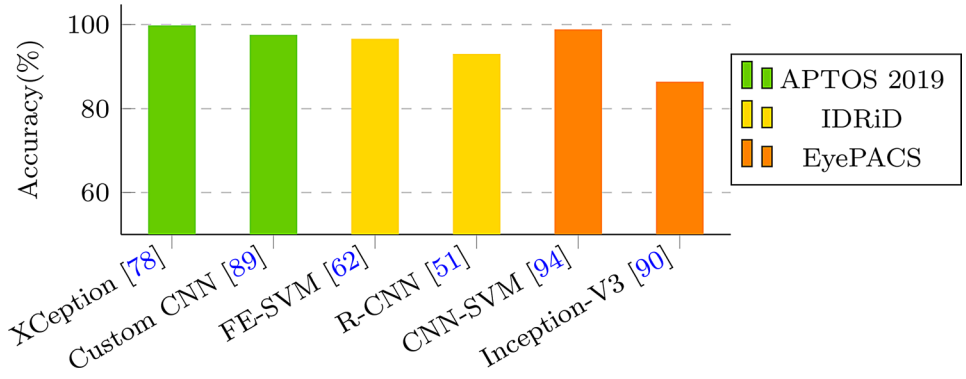


Fig. 14 Summary of the top two models for binary classification of DR across each dataset

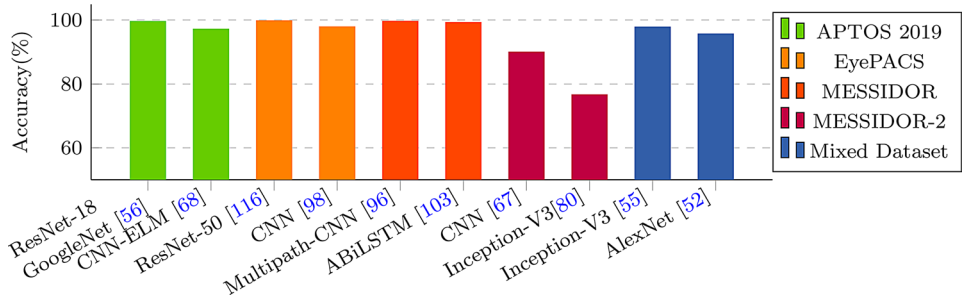


Fig. 15 Summary of the top two models for DR grading across each dataset

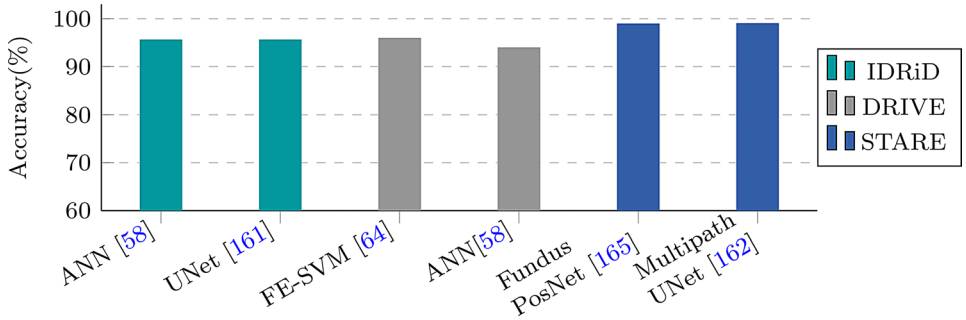


Fig. 16 Summary of the top two models for DR segmentation across each dataset

UNet [161], FE-SVM [64], FundusPosNet [165], and Multipath R UNet [162] showed good performance on each dataset, as highlighted in Fig. 16, despite their computational burden.

Several studies have shown that DL techniques are promising for creating effective systems for tasks like segmentation, localization, and grading in the diagnosis of DR. Various DL models and architectures, including pre-trained CNN, custom-designed CNN, ensemble models, and capsule networks, have been investigated. The reported results in the literature demonstrate their effectiveness in supporting DR diagnosis.

In addition, the pre-processing techniques used to analyze retinal fundus images have a significant impact on improving the performance of CNN models. A study conducted by Khaled, et al. [114] and Barhate, et al. [113], Yaqoob, et al. [123] and Martinez, et al. [120], and Alshahrani, et al. [56] and Butt, et al. [130] have utilized similar CNN architecture on the same dataset but have reported varying results. This indicates that the pre-processing techniques utilized have affected the model's performance. Given the crucial role of pre-processing techniques in model performance, we have summarized techniques employed in each reviewed papers in Tables 6, 7, 8, 9, 10. Additionally, the evolving research of DR grading is witnessing huge efforts in refining image pre-processing techniques.

As ensembling multiple DL models and capsule networks is resource-intensive, the researchers opted to stick with transfer learning techniques and a custom-designed

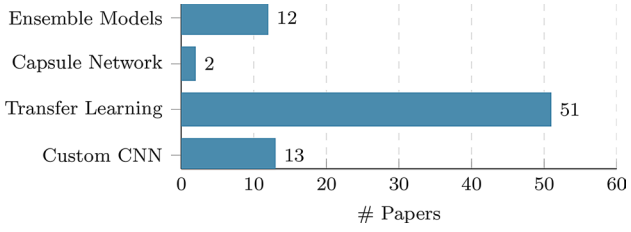


Fig. 17 Distribution of techniques employed in the reviewed scholarly papers

CNN to refine and build a lighter model. This is why transfer learning is also widely employed in DL techniques, as shown in Fig. 17. Figure 18 shows the current trends of DL techniques employed in the scholarly papers reviewed in this systematic literature review. Many transfer learning methods use ImageNet weights as initial weights; however, RETFound (self supervised learning based foundation model for retinal image) has been introduced as a foundation model designed for generalizable disease detection from retinal images [201]. This model is trained on 1.6 million unlabeled retinal images using self-supervised learning, allowing it to adapt efficiently to various clinical tasks with fewer labeled data. RETFound consistently outperforms traditional models in diagnosing sight-threatening eye diseases. On the other hand, integrating attention techniques into these architectures has shown encouraging results. Attention mechanisms, such as residual, spatial, and channel attention, help the model capture details from minor features or lesions. Although some researchers have used classical

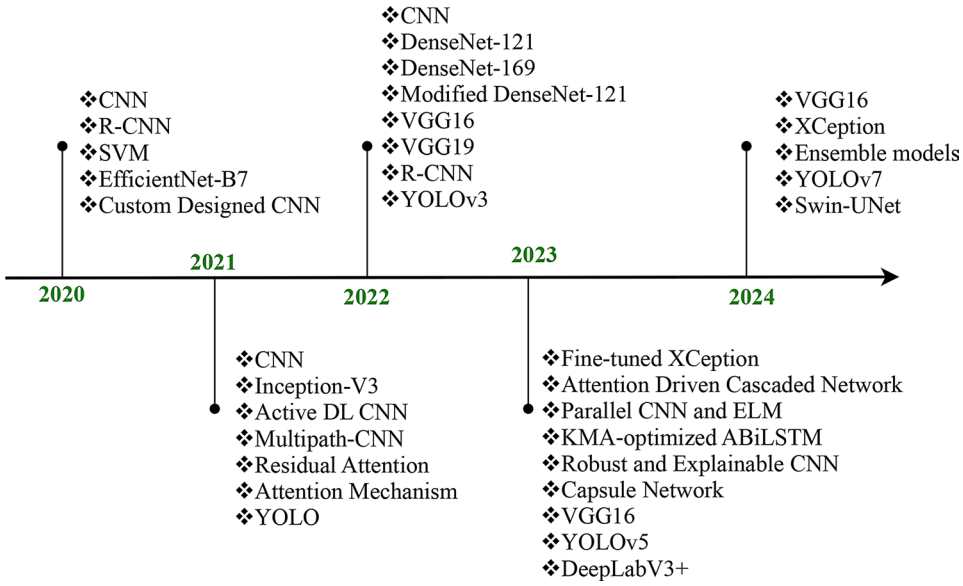


Fig. 18 Trends of deep learning models from the reviewed scholary papers

machine learning classifier algorithms for classification and employed lighter CNN models for feature extraction, they have also shown relatively good results. Furthermore, FL and explainable AI are gaining considerable attention in DR screening, addressing problems that hinder DL techniques, such as computational resource limitations and the black-box nature of the models.

Challenges and future directions

While ML methods have undoubtedly made significant strides in the realms of DR localization, grading, and segmentation, challenges persist, particularly when applying ML models to retinal fundus images. These challenges necessitate further attention and resolution to enhance the efficacy and reliability of ML-based approaches in this domain.

The shortage of high-quality labeled datasets remains a critical challenge, as the performance of ML models heavily depends on both the quality and quantity of annotated data. Obtaining a sufficient amount of accurately labeled data is particularly difficult due to the complex and time-intensive processes involved in retinal fundus image acquisition, filtering, annotation, and grading. Publicly available datasets often suffer from issues such as class imbalance, inseparability, and insufficient sample sizes, limiting the development of robust ML models. Furthermore, the small size and tiny nature of lesions, such as microaneurysms, hemorrhages, hard exudates, and cotton-wool spots, pose significant challenges for accurate segmentation and classification. These lesions often exhibit high intra-class variation and low inter-class contrast, making them difficult for models to distinguish. To advance research in this domain, it is essential to develop and provide access to high-resolution, well-annotated datasets for the research community.

ML models are resource-intensive, with many requiring significant computational resources for training and deployment. We suggest creating a lightweight model by incorporating attention mechanisms and utilizing model optimization techniques such as pruning, sparsifying, and knowledge distillation. This will enable deployment on edge devices such as mobile and IoT devices. Additionally, FL is expected to play a significant role in addressing resource constraints, particularly by enabling decentralized training without compromising data privacy. Its outcomes have shown promising potential in real-world applications, especially in fields like healthcare and finance. XAI will also be crucial in making AI models more understandable and interpretable, fostering trust and broader adoption in critical domains. However, XAI remains an active area of research, particularly in enabling AI models to accurately identify and explain the specific lesions associated with a disease. Achieving these capabilities would not only enhance diagnostic accuracy but also provide clinicians with actionable insights to improve patient care.

The study showcased the application of several assessment measures in reporting quantitative outcomes, indicating their capacity to cover many aspects of performance. However, a considerable number of ML methods assessed in this review have not been tested in clinical settings, raising doubts about their optimal performance in practical applications. Close collaboration between data scientists and ophthalmologists is crucial for ensuring the effective adoption of ML algorithms in clinical

settings. Additionally, it enhances the reliability and efficacy of ML algorithms in clinical settings.

Conclusions

This systematic review explores the challenges and opportunities in using ML for DR diagnosis from retinal fundus images. It identifies several datasets, though issues such as imbalanced data, small sample sizes, and inconsistent grading pose challenges. Various preprocessing techniques were reviewed, and common ML methods for DR localization, grading, and segmentation were analyzed. Key obstacles highlighted include low data quality, model complexity, and the lack of clinical validation of ML models.

Moving forward, ML models for DR screening need to be tailored for clinical settings, with lightweight models introduced to reduce computational complexity. FL enable collaborative model training across multiple institutions while preserving data privacy and improving dataset diversity. Additionally, high-quality, well-curated datasets should be collected and made accessible to the research community to improve model performance. Additionally, the involvement of ophthalmologists in defining use cases and further validation in real-world clinical settings will be essential to achieving accurate, ophthalmologist-level grading for DR diagnosis.

Abbreviations

ABiLSTM	Attention-based Bi-directional Long Short-term Memory
AHE	Adaptive Histogram Equalization
CLAHE	Contrast Limited Adaptive Histogram Equalization
CNN	Convolutional Neural Network
DL	Deep Learning
DME	Diabetic Macular Edema
DR	Diabetic Retinopathy
DWT	Discrete Wavelet Transform
FCH	Fuzzy Color Histogram
FE	Feature Extraction
FL	Federated Learning
FOV	Field of View
GA	Genetic Algorithm
GNN	Graph Neural Networks
HOG	Histogram of Oriented Gradients
ICDR	International Clinical Diabetic Retinopathy
IDF	International Diabetes Federation
KMA	Komodo MLIPIR algorithm
LBP	Local Binary Pattern
LTP	Local Ternary Pattern
ML	Machine Learning
MLP	Multilayer Perceptron
MPA	Marine Predators Algorithm
NPDR	Non Proliferative Diabetic Retinopathy
PCA	Principal Component Analysis
PDR	Proliferative Diabetic Retinopathy
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
RQ	Research Question
RETFound	self supervised learning based foundation model for retinal image
SIFT	Scale-invariant Feature Transform
XAI	Explainable Artificial Intelligence

Supplementary Information

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Supplementary Material 1

Authors contributions

All authors made a significant contribution to the work reported. F.M. contributed to article search, screening, and manuscript writing; F.M., T.G., Y.M., and D.W. did manuscript revisions and editions; T.G., Y.M., D.W., and F.S. did manuscript validations. All authors contributed to the approval of the published version and agreed on the journal to which the article was submitted.

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Data availability

The datasets used and/or analysed during the current study are available from the corresponding or first author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

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Competing interests

The authors declare that they have no competing interests.

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References

1. Antonetti DA, Klein R, Gardner TW. Diabetic retinopathy. N Engl J Med. 2012;366(13):1227–39. https://doi.org/10.1056/nejmra1005073.
2. Atlas D, et al. International diabetes federation. IDF diabetes atlas, 7th edn. Brussels, Belgium: International Diabetes Federation. 2015;33(2).
3. Yau JW, Rogers SL, Kawasaki R, Lamoureux EL, Kowalski JW, Bek T, Chen S-J, Dekker JM, Fletcher A, Grauslund J, et al. Global prevalence and major risk factors of diabetic retinopathy. Diabetes Care. 2012;35(3):556–64. https://doi.org/10.2337/dc11-1909.
4. Resnikoff S, Lansingh VC, Washburn L, Felch W, Gauthier T-M, Taylor HR, Eckert K, Parke D, Wiedemann P. Estimated number of ophthalmologists worldwide (international council of ophthalmology update): Will we meet the needs? Br J Ophthalmol. 2020;104(4):588–92. https://doi.org/10.1136/bjophthalmol-2019-314336.
5. Dean WH, Buchan JC, Gichuhi S, Faal H, Mpyet C, Resnikoff S, Gordon I, Matende I, Samuel A, Visser L, et al. Ophthalmology training in sub-saharan africa: A scoping review. Eye. 2021;35(4):1066–83. https://doi.org/10.1038/s41433-020-01335-7.
6. Wang SY, Andrews CA, Gardner TW, Wood M, Singer K, Stein JD. Ophthalmic screening patterns among youths with diabetes enrolled in a large us managed care network. JAMA Ophthalmol. 2017;135(5):432–38. https://doi.org/10.1001/jamaophthalmol.2017.0089.
7. Feder RS, Olsen TW, Prum BE, Summers CG, Olson RJ, Williams RD, Musch DC. Comprehensive adult medical eye evaluation preferred practice pattern® guidelines. Ophthalmology. 2016;123(1):209–36. https://doi.org/10.1016/j.ophtha.2015.10.047.
8. Dubow M, Pinhas A, Shah N, Cooper RF, Gan A, Gentile RC, Hendrix V, Sulai YN, Carroll J, Chui TY, et al. Classification of human retinal microaneurysms using adaptive optics scanning light ophthalmoscope fluorescein angiography. Invest Ophthalmol & Visual Sci. 2014;55(3):1299–309. https://doi.org/10.1167/iovs.13-13122.

9. Murugesan N, Üstunkaya T, Feener EP. Thrombosis and hemorrhage in diabetic retinopathy: A perspective from an inflammatory standpoint. Semin Thromb Hemost. 2015;659–64. https://doi.org/10.1055/s-0035-1556731.
10. Amin J, Sharif M, Yasmin M, et al. A review on recent developments for detection of diabetic retinopathy. Scientifica. 2016;2016. https://doi.org/10.1155/2016/6838976.
11. Wilkinson CP, Ferris FL III, Klein RE, Lee PP, Agardh CD, Davis M, Dills D, Kampik A, Pararajasegaram R, Verdaguer JT, et al. Proposed international clinical diabetic retinopathy and diabetic macular edema disease severity scales. Ophthalmology. 2003;110(9):1677–82. https://doi.org/10.1016/S0161-6420(03)00475-5.
12. Mateen M, Junhao W, Nasrullah N, Song S, Shaukat H, et al. Exudate detection for diabetic retinopathy using pretrained convolutional neural networks. Complexity. 2020;(9):5801870.
13. Shah A, Clarida W, Amelon R, Hernaez-Ortega MC, Navea A, Morales- Olivas J, Dolz-Marco R, Verbraak F, Jorda PP, Heijden AA, et al. Validation of automated screening for referable diabetic retinopathy with an autonomous diagnostic artificial intelligence system in a Spanish population. J Diabetes Sci Technol. 2021;15(3):655–63. https://doi.org/10.1177/1932296820906212.
14. Ishtiaq U, Abdul Kareem S, Abdullah ERMF, Mujtaba G, Jahangir R, Ghafoor HY. Diabetic retinopathy detection through artificial intelligent techniques: A review and open issues. Multimedia Tools Appl. 2020;79:15209–52. https://doi.org/10.1007/s11042-018-7044-8.
15. Lakshminarayanan V, Kheradfallah H, Sarkar A, Jothi Balaji J. Automated detection and diagnosis of diabetic retinopathy: A comprehensive survey. J Imag. 2021;7(9):165. https://doi.org/10.3390/jimaging7090165.
16. Nadeem MW, Goh HG, Hussain M, Liew S-Y, Andonovic I, Khan MA. Deep learning for diabetic retinopathy analysis: A review, research challenges, and future directions. Sensors. 2022;22(18):6780. https://doi.org/10.3390/s22186780.
17. Kárason KT, Vo D, Grauslund J, Rasmussen ML. Comparison of different methods of retinal imaging for the screening of diabetic retinopathy: A systematic review. Acta Ophthalmologica. 2022;100(2):127–35. https://doi.org/10.1111/aos.14767.
18. Islam MM, Yang H-C, Poly TN, Jian W-S, Li Y-CJ. Deep learning algorithms for detection of diabetic retinopathy in retinal fundus photographs: A systematic review and meta-analysis. Comput Met Programs Biomed. 2020;191:105320. https://doi.org/10.1016/j.cmpb.2020.105320.
19. Ilesanmi AE, Ilesanmi T, Gbotoso AG. A systematic review of retinal fundus image segmentation and classification methods using convolutional neural networks. Healthcare Analytics. 2023;100261. https://doi.org/10.1016/j.health.2023.100261.
20. Sebastian A, Elharrouss O, Al-Maadeed S, Almaadeed N. A survey on deeplearning-based diabetic retinopathy classification. Diagnostics. 2023;13(3):345. https://doi.org/10.3390/diagnostics13030345.
21. Tsiknakis N, Theodoropoulos D, Manikis G, Ktistakis E, Boutsora O, Berto A, Scarpa F, Scarpa A, Fotiadis DI, Marias K. Deep learning for diabetic retinopathy detection and classification based on fundus images: A review. Comput Biol Med. 2021;135:104599. https://doi.org/10.1016/j.compbiomed.2021.104599.
22. Selvachandran G, Quek SG, Paramesran R, Ding W, Son LH. Developments in the detection of diabetic retinopathy: A state-of-the-art review of computer-aided diagnosis and machine learning methods. Artif Intell Rev. 2023;56(2):915–64. https://doi.org/10.1007/s10462-022-10185-6.
23. Alyoubi WL, Shalash WM, Abulkhair MF. Diabetic retinopathy detection through deep learning techniques: A review. Inf Med Unlocked. 2020;20:100377. https://doi.org/10.1016/j.imu.2020.100377.
24. Chu A, Squirrell D, Phillips AM, Vaghefi E. Essentials of a robust deep learning system for diabetic retinopathy screening: A systematic literature review. J Ophthalmol. 2020;2020(1):8841927. https://doi.org/10.1155/2020/8841927.
25. Grzybowski A, Brona P, Lim G, Ruamviboonsuk P, Tan GS, Abramoff M, Ting DS. Artificial intelligence for diabetic retinopathy screening: A review. Eye. 2020;34(3):451–60. https://doi.org/10.1038/s41433-019-0566-0.
26. Wewetzer L, Held LA, Steinhäuser J. Diagnostic performance of deeplearning-based screening methods for diabetic retinopathy in primary care—a meta-analysis. PLoS One. 2021;16(8):0255034. https://doi.org/10.1371/journal.pone.0255034.
27. Wu J-H, Liu TA, Hsu W-T, Ho JH-C, Lee C-C. Performance and limitation of machine learning algorithms for diabetic retinopathy screening: Meta-analysis. J Med Internet Res. 2021;23(7):23863. https://doi.org/10.2196/23863.
28. Brennan SE, Munn Z. Prisma 2020: A reporting guideline for the next generation of systematic reviews. JBI Evidence Synth. 2021;19(5):906–08. https://doi.org/10.11124/JBIES-21-00112.
29. Karthik SD. Maggie: APTOS 2019 blindness detection. Kaggle. Accessed: June 9, 2025;2019. https://kaggle.com/competitions/aptos2019-blindness-detection.
30. Decencière E, Zhang X, Cazuguel G, Lay B, Cochener B, Trone C, Gain P, Ordonez R, Massin P, Erginay A, et al. Feedback on a publicly distributed image database: The messidor database. Image Anal And Stereol. 2014;33(3):231–34. https://doi.org/10.5566/ias.1155.
31. Abràmoff MD, Folk JC, Han DP, Walker JD, Williams DF, Russell SR, Massin P, Cochener B, Gain P, Tang L, et al. Automated analysis of retinal images for detection of referable diabetic retinopathy. JAMA Ophthalmol. 2013;131(3):351–57. https://doi.org/10.1001/jamaophthalmol.2013.1743.
32. Kauppi T, Kalesnykiene V, Kamarainen J-K, Lensu L, Sorri I, Uusitalo H, Kälviäinen H, Pietilä J. Diaretdb0: evaluation database and methodology for diabetic retinopathy algorithms. In: Machine vision and pattern recognition research group. Vol. 73. Finland: Lappeenranta University of Technology; 2006. p. 1–17. https://doi.org/10.5244/C.21.15.
33. Kauppi T, Kalesnykiene V, Kamarainen J-K, Lensu L, So Rri I, Raninen A, Voutilainen R, Uusitalo H, Kälviäinen H, Pietilä J. The diaretdb1 diabetic retinopathy database and evaluation protocol. BMVC. 2007;1:10. https://doi.org/10.5244/C.21.15.
34. Li T, Gao Y, Wang K, Guo S, Liu H, Kang H. Diagnostic assessment of deep learning algorithms for diabetic retinopathy screening. Inf Sci (Ny). 2019;501:511–22. https://doi.org/10.1016/j.ins.2019.06.011.
35. Dugas E, et al. Diabetic retinopathy detection. Kaggle; 2015 [cited 2025 Jun 9]. https://kaggle.com/competitions/diabetic-retinopathy-detection.
36. DRIVE: Digital retinal images for vessel extraction. Accessed: June 9, 2025. https://drive.grand-challenge.org/.
37. STructured analysis of the retina. Accessed: June 9, 2025. https://cecas.clemson.edu/~ahoover/stare/.
38. CHASE DB1. Retinal dataset. Accessed: June 9, 2025;2011. http://sec.kingston.ac.uk/.
39. Porwal P, Pachade S, Kamble R, Kokare M, Deshmukh G, Sahasrabuddhe V, Meriaudeau F. Indian Diabetic Retinopathy Image Dataset (IDRiD). IEEE Dataport. Accessed: June. 2018;9:2025. https://doi.org/10.21227/H25W98.
40. Kumar V, Paul K. Fundus imaging-based healthcare: Present and future. ACM Trans Comput Healthc. 2023;4(3):1–34. https://doi.org/10.1145/3586580.
41. Lepetit-Aimon G, Playout C, Boucher M, Duval R, Brent M, Cheriet F. MAPLES-DR: MESSIDOR anatomical and pathological labels for explainable screening of diabetic retinopathy. Jan 2024. https://doi.org/10.1038/s41597-024-03739-6.
42. Zhang X, Thibault G, Decencière E, Marcotegui B, Laÿ B, Danno R, Cazuguel G, Quellec G, Lamard M, Massin P, et al. Exudate detection in color retinal images for mass screening of diabetic retinopathy. Med Image Anal. 2014;18(7):1026–43. https://doi.org/10.1016/j.media.2014.05.004.
43. Niemeijer M, Van Ginneken B, Cree MJ, Mizutani A, Quellec G, Sánchez CI, Zhang B, Hornero R, Lamard M, Muramatsu C, et al. Retinopathy online challenge: Automatic detection of microaneurysms in digital color fundus photographs. IEEE Trans Med Imag. 2009;29(1):185–95. https://doi.org/10.1109/TMI.2009.2033909.
44. Raj A, Tiwari AK, Martini MG. Fundus image quality assessment: Survey, challenges, and future scope. IET Image Process. 2019;13(8):1211–24. https://doi.org/10.1049/iet-ipr.2018.6212.
45. Yan YN, Wang YX, snmXu L, Xu J, Wei WB, Jonas JB. Fundus tessellation: Prevalence and associated factors: The beijing eye study 2011. Ophthalmology. 2015;122(9):1873–80.
46. Bindhya P, Jegan C, Raj V. A review on methods of enhancement and denoising in retinal fundus images. Int J Mult Comp Sci Eng. 2020;8:1–9. https://doi.org/10.26438/ijcse/v8i12.19.
47. Bala AA, Priya PA, Maik V. Retinal image enhancement using curvelet based sigmoid mapping of histogram equalization. J Phys: Conf Series. IOP Publishing; 2021;1964:062034. https://doi.org/10.1088/1742-6596/1964/6/062034.
48. Sahu S, Singh AK, Ghrera S, Elhoseny M, et al. An approach for de-noising and contrast enhancement of retinal fundus image using clahe. Opt Laser Technol. 2019;110:87–98. https://doi.org/10.1016/j.optlastec.2018.06.061.
49. Wu B, Zhu W, Shi F, Zhu S, Chen X. Automatic detection of microaneurysms in retinal fundus images. Computerized Med Imag Graphics. 2017;55:106–12. https://doi.org/10.1016/j.compmedimag.2016.08.001.

50. Wang J, Li Y-J, Yang K-F. Retinal fundus image enhancement with image decomposition and visual adaptation. Comput Biol Med. 2021;128:104116. https://doi.org/10.1016/j.compbiomed.2020.104116.
51. Ghan G, Chavan S, Chaudhari A. Diabetic retinopathy classification using deep learning. 2020 Fourth International Conference on Inventive Systems and Control (ICISC). IEEE; 2020;761–65. https://doi.org/10.1109/ICISC47916.2020.9171139.
52. Vaishnavi J, Ravi S, Anbarasi A. An efficient adaptive histogram based segmentation and extraction model for the classification of severities on diabetic retinopathy. Multimedia Tools Appl. 2020;79:30439–52. https://doi.org/10.1007/s11042-020-09288-5.
53. Dayana AM, Emmanuel WS. Deep learning enabled optimized feature selection and classification for grading diabetic retinopathy severity in the fundus image. Neural Comput Appl. 2022;34(21):18663–83. https://doi.org/10.1007/s00521-022-07471-3.
54. Saranya P, Umamaheswari K. Detection of exudates from retinal images for non-proliferative diabetic retinopathy detection using deep learning model. Multimedia Tools Appl. 2023;1–21. https://doi.org/10.1007/s11042-023-17462-8.
55. Bilal A, Zhu L, Deng A, Lu H, Wu N. Ai-based automatic detection and classification of diabetic retinopathy using u-net and deep learning. Symmetry. 2022;14(7):1427. https://doi.org/10.3390/sym14071427.
56. Alshahrani M, Al-Jabbar M, Senan EM, Ahmed IA, Saif JAM. Hybrid methods for fundus image analysis for diagnosis of diabetic retinopathy development stages based on fusion features. Diagnostics. 2023;13(17):2783. https://doi.org/10.3390/diagnostics13172783.
57. Gargari MS, Seyedi MH, Alilou M. Segmentation of retinal blood vessels using u-net++ architecture and disease prediction. Electronics. 2022;11(21):3516. https://doi.org/10.3390/electronics11213516.
58. Toptaş B, Hanbay D. Retinal blood vessel segmentation using pixel-based feature vector. Biomed Signal Process Control. 2021;70:103053. https://doi.org/10.1016/j.bspc.2021.103053.
59. Tamim N, Elshrkawey M, Abdel Azim G, Nassar H. Retinal blood vessel segmentation using hybrid features and multi-layer perceptron neural networks. Symmetry. 2020;12(6):894. https://doi.org/10.3390/sym12060894.
60. Tchinda BS, Tchiotsop D, Noubom M, Louis-Dorr V, Wolf D. Retinal blood vessels segmentation using classical edge detection filters and the neural network. Inf Med Unlocked. 2021;23:100521. https://doi.org/10.1016/j.imu.2021.100521.
61. Skouta A, Elmoufidi A, Jai-Andaloussi S, Ouchetto O. Hemorrhage semantic segmentation in fundus images for the diagnosis of diabetic retinopathy by using a convolutional neural network. J Big Data. 2022;9(1):1–24. https://doi.org/10.1186/s40537-022-00632-0.
62. Bibi I, Mir J, Raja G. Automated detection of diabetic retinopathy in fundus images using fused features. Phys Eng Sci Med. 2020;43(4):1253–64. https://doi.org/10.1007/s13246-020-00929-5.
63. Athira T, Nair JJ. Diabetic retinopathy grading from color fundus images: An autotuned deep learning approach. Procedia Comput Sci. 2023;218:1055–66. https://doi.org/10.1016/j.procs.2023.01.085.
64. Sidhu R, Sachdeva J, Katoch D. Segmentation of retinal blood vessels by a novel hybrid technique-principal component analysis (pca) and contrast limited adaptive histogram equalization (clahe). Microvasc Res. 2023;148:104477. https://doi.org/10.1016/j.mvr.2023.104477.
65. Huang Y, Zhong Z, Yuan J, Tang X. Efficient and robust optic disc detection and fovea localization using region proposal network and cascaded network. Biomed Signal Process Control. 2020;60:101939. https://doi.org/10.1016/j.bspc.2020.101939.
66. Zuiderveld K. Contrast limited adaptive histogram equalization. Graphics Gems. 1994;474–85.
67. Ashwini K, Dash R. Grading diabetic retinopathy using multiresolution based cnn. Biomed Signal Process Control. 2023;86:105210. https://doi.org/10.1016/j.bspc.2023.105210.
68. Nahiduzzaman M, Islam MR, Goni MOF, Anower MS, Ahsan M, Haider J, Kowalski M. Diabetic retinopathy identification using parallel convolutional neural network based feature extractor and elm classifier. Expert Syst With Applications. 2023;217:119557. https://doi.org/10.1016/j.eswa.2023.119557.
69. Bhardwaj C, Jain S, Sood M. Diabetic retinopathy severity grading employing quadrant-based inception-v3 convolution neural network architecture. Int J Imag Syst And Technol. 2021;31(2):592–608. https://doi.org/10.1002/ima.22510.
70. Jabbar MK, Yan J, Xu H, Ur Rehman Z, Jabbar A. Transfer learningbased model for diabetic retinopathy diagnosis using retinal images. Brain Sci. 2022;12(5):535. https://doi.org/10.3390/brainsci12050535.
71. Kasim Ö. Ensemble classification based optimized transfer learning feature method for early stage diagnosis of diabetic retinopathy. J Ambient Intel Humanized Comput. 2023;1–12. https://doi.org/10.1007/s12652-023-04648-z.
72. Kommaraju R, Anbarasi M. Diabetic retinopathy detection using convolutional neural network with residual blocks. Biomed Signal Process Control. 2024;87:105494. https://doi.org/10.1016/j.bspc.2023.105494.
73. Kalyani G, Janakiramaiah B, Karuna A, Prasad LN. Diabetic retinopathy detection and classification using capsule networks. Complex Intell Syst. 2023;9(3):2651–64. https://doi.org/10.1007/s40747-021-00318-9.
74. Novitasari DCR, Fatmawati F, Hendradi R, Rohayani H, Nariswari R, Arnita A, Hadi MI, Saputra RA, Primadewi A. Image fundus classification system for diabetic retinopathy stage detection using hybrid cnn-delm. Big Data Cogn. Comput. 2022;6(4):146. https://doi.org/10.3390/bdcc6040146.
75. Huang S, Li J, Xiao Y, Shen N, Xu T. Rtnet: Relation transformer network for diabetic retinopathy multi-lesion segmentation. IEEE Trans Med Imag. 2022;41(6):1596–607. https://doi.org/10.1109/TMI.2022.3143833.
76. Xu Y, Zhou Z, Li X, Zhang N, Zhang M, Wei P, et al. Ffu-net: Feature fusion u-net for lesion segmentation of diabetic retinopathy. BioMed Res Int. 2021;2021. https://doi.org/10.1155/2021/6644071.
77. Guo Y, Peng Y. Carnet: Cascade attentive refinenet for multi-lesion segmentation of diabetic retinopathy images. Complex Intell Syst. 2022;8(2):1681–701. https://doi.org/10.1007/s40747-021-00630-4.
78. Beghriche T, Attallah B, Brik Y, Djerioui M. A multi-level fine-tuned deep learning based approach for binary classification of diabetic retinopathy. Chemom And Intell Lab Syst. 2023;237:104820. https://doi.org/10.1016/j.chemolab.2023.104820.
79. Cao J, Chen J, Zhang X, Peng Y. Diabetic retinopathy classification based on dense connectivity and asymmetric convolutional neural network. Neural Comput Appl. 2022;1–14. https://doi.org/10.1007/s00521-022-07952-5.
80. Bhardwaj C, Jain S, Sood M. Transfer learning based robust automatic detection system for diabetic retinopathy grading. Neural Comput Appl. 2021;33(20):13999–4019. https://doi.org/10.1007/s00521-021-06042-2.
81. Haddad RA, Akansu AN, et al. A class of fast gaussian binomial filters for speech and image processing. IEEE Trans Signal Process. 1991;39(3):723–27. https://doi.org/10.1109/78.80892.
82. Huang T, Yang G, Tang G. A fast two-dimensional median filtering algorithm. IEEE Trans Acoust Speech Signal Process. 1979;27(1):13–18. https://doi.org/10.1109/TASSP.1979.1163188.
83. Alessio SM. DiscreteWavelet transform (DWT). Cham: Springer; 2016. p. 645–714. https://doi.org/10.1007/978-3-319-25468-5_14.
84. De Siqueira FR, Schwartz WR, Pedrini H. Multi-scale gray level cooccurrence matrices for texture description. Neurocomputing. 2013;120:336–45. https://doi.org/10.1016/j.neucom.2012.09.042.
85. Dalal N, Triggs B. Histograms of oriented gradients for human detection. 2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR'05). 2005;1:886–93. https://doi.org/10.1109/CVPR.2005.177.Ieee.
86. Salam AA, Mahadevappa M, Das A, Nair MS. Drg-net: A graph neural network for computer-aided grading of diabetic retinopathy. Signal, Image Video Process. 2022;16(7):1869–75. https://doi.org/10.1007/s11760-022-02146-x.
87. Chetoui M, Akhloufi MA. Explainable diabetic retinopathy using efficientnet. 2020 42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC). 2020, pp. 1966–69. https://doi.org/10.1109/EMBC44109.2020.9175664.
88. Jiang H, Xu J, Shi R, Yang K, Zhang D, Gao M, Ma H, Qian W. A multi-label deep learning model with interpretable grad-cam for diabetic retinopathy classification. 2020 42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC). 2020:1560–63. https://doi.org/10.1109/EMBC44109.2020.9175884.
89. Rêgo S, Dutra-Medeiros M, Soares F, Monteiro-Soares M. Screening for diabetic retinopathy using an automated diagnostic system based on deep learning: Diagnostic accuracy assessment. Ophthalmologica. 2021;244(3):250–57. https://doi.org/10.1159/000512638.
90. Gurcan OF, Beyca ÖF, Dogan. O. A comprehensive study of machine learning methods on diabetic retinopathy classification. Int J Comput Intell Syst. 2021;14(1):1132–41.
91. Anoop B, et al. Binary classification of dr-diabetic retinopathy using cnn with fundus colour images. Materials Today: Proceedings. 2022;58:212–16. https://doi.org/10.1016/j.matpr.2022.01.466.

92. Bala R, Sharma A, Goel N. A novel convolutional neural network architecture for diabetic retinopathy screening. 2022 4th International Conference on Artificial Intelligence and Speech Technology (AIST). IEEE. 2022:1–6. https://doi.org/10.1109/AIST55798.2022.10065181.
93. Nandakumar R, Saranya P, Ponnusamy V, Hazra S, Gupta A. Detection of diabetic retinopathy from retinal images using densenet models. Comput Syst Sci Eng. 2023;45(1). https://doi.org/10.32604/csse.2023.028703.
94. Ishtiaq U, Abdullah ERMF, Ishtiaque Z. A hybrid technique for diabetic retinopathy detection based on ensemble-optimized cnn and texture features. Diagnostics. 2023;13(10):1816. https://doi.org/10.3390/diagnostics13101816.
95. Huang G, Liu Z, Van Der Maaten L, Weinberger KQ. Densely connected convolutional networks. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. 2017, pp. 4700–08. https://doi.org/10.1109/CVPR.2017.243.
96. Gayathri S, Gopi VP, Palanisamy P. Diabetic retinopathy classification based on multipath cnn and machine learning classifiers. Phys Eng Sci Med. 2021;44(3):639–53. https://doi.org/10.1007/s13246-021-01012-3.
97. Yue G, Li Y, Zhou T, Zhou X, Liu Y, Wang T. Attention-driven cascaded network for diabetic retinopathy grading from fundus images. Biomed Signal Process Control. 2023;80:104370. https://doi.org/10.1016/j.bspc.2022.104370.
98. Qureshi I, Ma J, Abbas Q. Diabetic retinopathy detection and stage classification in eye fundus images using active deep learning. Multimedia Tools Appl. 2021;80:11691–721. https://doi.org/10.1007/s11042-020-10238-4.
99. Liu P, Wang L, Ranjan R, He G, Zhao L. A survey on active deep learning: From model driven to data driven. ACM Comput Surv (CSUR). 2022;54(10s):1–34. https://doi.org/10.1145/3510414.
100. Koch G, Zemel R, Salakhutdinov R, et al. Siamese neural networks for one-shot image recognition. ICML Deep Learning Workshop. Lille. 2015:2.
101. Deepa V, Sathish Kumar C, Cherian T. Automated grading of diabetic retinopathy using cnn with hierarchical clustering of image patches by siamese network. Phys Eng Sci Med. 2022;45(2):623–35. https://doi.org/10.1007/s13246-022-01129-z.
102. Hossen MS, Reza AA, Mishu MC. An automated model using deep convolutional neural network for retinal image classification to detect diabetic retinopathy. Proceedings of the International Conference on Computing Advancements. 2020, pp. 1–8. https://doi.org/10.1145/3377049.3377067.
103. Fan R, Liu Y, Zhang R. Multi-scale feature fusion with adaptive weighting for diabetic retinopathy severity classification. Electronics. 2021;10(12):1369. https://doi.org/10.3390/electronics10121369.
104. Sikder N, Masud M, Bairagi AK, Arif ASM, Nahid A-A, Alhumyani HA. Severity classification of diabetic retinopathy using an ensemble learning algorithm through analyzing retinal images. Symmetry. 2021;13(4):670.
105. Abirami A, Kavitha R. A novel automated komodo mlipir optimization-based attention bilstm for early detection of diabetic retinopathy. Signal, Image Video Process. 2023;1–9. https://doi.org/10.1007/s11760-022-02407-9.
106. Mercaldo F, Di Giammarco M, Apicella A, Di Iadarola G, Cesarelli M, Martinelli F, Santone A. Diabetic retinopathy detection and diagnosis by means of robust and explainable convolutional neural networks. Neural Comput Appl. 2023;1–13. https://doi.org/10.1007/s00521-023-08608-8.
107. Suyanto S, Ariyanto AA, Ariyanto AF. Komodo mlipir algorithm. Appl Soft Comput. 2022;114:108043. https://doi.org/10.1016/j.asoc.2021.108043.
108. Gu L, Li T, He J. Classification of diabetic retinopathy grade based on g-enet convolutional neural network model: Convolutional neural networks are used to solve the problem of diabetic retinopathy grade classification. Proceedings of the 2023 7th International Conference on Electronic Information Technology and Computer Engineering. 2023, pp. 1590–94. https://doi.org/10.1145/3650400.36506.
109. Xiao Z, Zhang Y, Wu J, Zhang X. Se-midnet based on deep learning for diabetic retinopathy classification. Proceedings of the 2021 7th International Conference on Computing and Artificial Intelligence. 2021, pp. 92–98. https://doi.org/10.1145/3467707.3467720.
110. Santos M, Valadao C, Resende C, Cavalieri D. Predicting diabetic retinopathy stage using siamese convolutional neural network. Comput Methods Biomech Biomed Eng: Imag Visual. 2024;12(1):2297017. https://doi.org/10.1080/21681163.2023.2297017.
111. Scarselli F, Gori M, Tsoi AC, Hagenbuchner M, Monfardini G. The graph neural network model. IEEE Trans On Neural Networks. 2008;20(1):61–80. https://doi.org/10.1109/TNN.2008.2005605.
112. Onjun R, Lowmunkhong S, Kaennakham S, Phattaramarut K. Diabetic retinopathy detection using a convolutional neural network enhanced by wavelet transformation pooling. Proceedings of the 2024 16th International Conference on Bioinformatics and Biomedical Technology. 2024:265–69. https://doi.org/10.1145/3674658.3674699.
113. Barhate N, Bhave S, Bhise R, Sutar RG, Karia DC. Reducing overfitting in diabetic retinopathy detection using transfer learning. 2020 IEEE 5th International Conference on Computing Communication and Automation (ICCCA). IEEE. 2020:298–301. https://doi.org/10.1109/ICCCA49541.2020.9250772.
114. Khaled O, El-Sahhar M, El-Dine MA, Talaat Y, Hassan YM, Hamdy A. Cascaded architecture for classifying the preliminary stages of diabetic retinopathy. Proceedings of the 9th International Conference on Software and Information Engineering. 2020, pp. 108–12. https://doi.org/10.1145/3436829.3436854.
115. Khan Z, Khan FG, Khan A, Rehman ZU, Shah S, Qummar S, Ali F, Pack S. Diabetic retinopathy detection using vgg-nin a deep learning architecture. IEEE Access. 2021;9:61408–16. https://doi.org/10.1109/ACCESS.2021.3074422.
116. Yaqoob MK, Ali SF, Kareem I, Fraz MM. Feature-based optimized deep residual network architecture for diabetic retinopathy detection. 2020 IEEE 23rd International Multitopic Conference (INMIC). IEE; 2020, pp. 1–6. https://doi.org/10.1109/INMIC50486.2020.9318096.
117. Saeed F, Hussain M, Aboalsamh HA. Automatic diabetic retinopathy diagnosis using adaptive fine-tuned convolutional neural network. IEEE Access. 2021;9:41344–59. https://doi.org/10.1109/ACCESS.2021.3065273.
118. Rajkumar R, Jagathishkumar T, Ragul D, Selvarani AG. Transfer learning approach for diabetic retinopathy detection using residual network. 2021 6th International Conference on Inventive Computation Technologies (ICICT). IEEE. 2021:1189–93. https://doi.org/10.1109/ICICT50816.2021.9358468.
119. Senbatu DH, Girma BS, Ayano YM. Tomato leaf disease detection and classification using custom modified alexnet. Pan African Conference on Artificial Intelligence. Springer; 2022, pp. 95–113. https://doi.org/10.1007/978-3-031-31327-1_6.
120. Martinez-Murcia FJ, Ortiz A, Ramırez J, Górriz JM, Cruz R. Deep residual transfer learning for automatic diagnosis and grading of diabetic retinopathy. Neurocomputing. 2021;452:424–34. https://doi.org/10.1016/j.neucom.2020.04.148.
121. Guo W, Guo R, Du X, Zhao Y, Li X, Toe TT. Diabetic retinopathy detection method based on improved convolutional neural network using fine-tuning. 2022 4th International Conference on Robotics and Computer Vision (ICRCV). 2022, pp. 113–17. https://doi.org/10.1109/ICRCV55858.2022.9953225.
122. Maaliw RR, Mabunga ZP, De Veluz MRD, Alon AS, Lagman AC, Garcia MB, Lacatan LL, Dellosa RM. An enhanced segmentation and deep learning architecture for early diabetic retinopathy detection. 2023 IEEE 13th Annual Computing and Communication Workshop and Conference (CCWC). IEEE. 2023:0168–75. https://doi.org/10.1109/CCWC57344.2023.10099069.
123. Yaqoob MK, Ali SF, Bilal M, Hanif MS, Al-Saggaf UM. Resnet based deep features and random forest classifier for diabetic retinopathy detection. Sensors. 2021;21(11):3883. https://doi.org/10.3390/s21113883.
124. Asia A-O, Zhu C-Z, Althubiti SA, Al-Alimi D, Xiao Y-L, Ouyang P-B, Al-Qaness MA. Detection of diabetic retinopathy in retinal fundus images using cnn classification models. Electronics. 2022;11(17):2740. https://doi.org/10.3390/electronics11172740.
125. Bhimavarapu U, Battineni G. Deep learning for the detection and classification of diabetic retinopathy with an improved activation function. Healthcare. MDPI. 2022;11:97. https://doi.org/10.3390/healthcare11010097.
126. Riaz H, Park J, Choi H, Kim H, Kim J. Deep and densely connected networks for classification of diabetic retinopathy. Diagnostics. 2020;10(1):24. https://doi.org/10.3390/diagnostics10010024.
127. Ayala A, Ortiz Figueroa T, Fernandes B, Cruz F. Diabetic retinopathy improved detection using deep learning. Appl Sci. 2021;11(24):11970. https://doi.org/10.3390/app112411970.
128. Raja Sarobin M, Panjanathan V, R. Diabetic retinopathy classification using cnn and hybrid deep convolutional neural networks. Symmetry. 2022;14(9):1932. https://doi.org/10.3390/sym14091932.
129. Zhang Z. Deep-learning-based early detection of diabetic retinopathy on fundus photography using efficientnet. Proceedings of the 2020 the 4th International Conference on Innovation in Artificial Intelligence. 2020, pp. 70–74. https://doi.org/10.1145/3390557.3394303.
130. Butt MM, Iskandar DA, Abdelhamid SE, Latif G, Alghazo R. Diabetic retinopathy detection from fundus images of the eye using hybrid deep learning features. Diagnostics. 2022;12(7):1607. https://doi.org/10.3390/diagnostics12071607.
131. Doshi N, Oza U, Kumar P. Diabetic retinopathy classification using downscaling algorithms and deep learning. 2020 7th International Conference on Signal Processing and Integrated Networks (SPIN). IEEE. 2020:950–55. https://doi.org/10.1109/SPIN48934.2020.9071423.

132. Sudarmadji PW, Pakan PD, Dillak RY. Diabetic retinopathy stages classification using improved deep learning. 2020 International Conference on Informatics, Multimedia, Cyber and Information System (ICIMCIS). IEEE. 2020:104–09. https://doi.org/10.1109/ICIMCIS51567.2020.9354281.
133. Dutta A, Agarwal P, Mittal A, Khandelwal S. Detecting grades of diabetic retinopathy by extraction of retinal lesions using digital fundus images. Res Biomed Eng. 2021;37:641–56. https://doi.org/10.1007/s42600-021-00177-w.
134. Lee C-H, Ke Y-H. Fundus images classification for diabetic retinopathy using deep learning. Proceedings of the 13th International Conference on Computer Modeling and Simulation. 2021, pp. 264–70. https://doi.org/10.1145/3474963.3475849.
135. Holland JH. Genetic algorithms. Sci Amer. 1992;267(1):66–73. https://doi.org/10.1038/scientificamerican0792-66.
136. Iandola FN, Han S, Moskewicz MW, Ashraf K, Dally WJ, Keutzer K. Squeezenet: Alexnet-level accuracy with 50x fewer parameters and¡ 0.5 mb model size. arXiv preprint arXiv:1602.07360. 2016. https://doi.org/10.48550/arXiv.1602.07360.
137. Parthiban K, Kamarasan M. Diabetic retinopathy detection and grading of retinal fundus images using coyote optimization algorithm with deep learning. Multimedia Tools Appl. 2023;82(12):18947–66. https://doi.org/10.1007/s11042-022-14234-8.
138. Pierezan J, Coelho LDS. Coyote optimization algorithm: A new metaheuristic for global optimization problems. 2018 IEEE Congress on Evolutionary Computation (CEC). 2018, pp. 1–8. https://doi.org/10.1109/CEC.2018.8477769.IEEE.
139. He K, Zhang X, Ren S, Sun J. Deep residual learning for image recognition. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. 2016, pp. 770–78. https://doi.org/10.1109/CVPR.2016.90.
140. Sunkari S, Sangam A, Suchetha M, Raman R, Rajalakshmi R, Tamilselvi S, et al. A refined resnet18 architecture with swish activation function for diabetic retinopathy classification. Biomed Signal Process Control. 2024;88:105630. https://doi.org/10.1016/j.bspc.2023.105630.
141. Howard AG, Zhu M, Chen B, Kalenichenko D, Wang W, Weyand T, Andreetto M, Adam H. Mobilenets: Efficient convolutional neural networks for mobile vision applications. arXiv preprint arXiv:1704.04861. 2017. https://doi.org/10.48550/arXiv.1704.04861.
142. Szegedy C, Liu W, Jia Y, Sermanet P, Reed S, Anguelov D, Erhan D, Vanhoucke V, Rabinovich A. Going deeper with convolutions. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. 2015:1–9. https://doi.org/10.1109/CVPR.2015.7298594.
143. Gangwar AK, Ravi V. Diabetic retinopathy detection using transfer learning and deep learning. In: Bhateja V, Peng S-L, Satapathy SC, Zhang Y-D, editors. Evolution in computational intelligence. Singapore: Springer; 2021. p. 679–89. https://doi.org/10.1007/978-981-15-5788-0_64.
144. Huang Z. Deep learning-based classification of diabetic retinopathy. Proceedings of the 2023 4th International Symposium on Artificial Intelligence for Medicine Science. 2023:371–75. https://doi.org/10.1145/3644116.3644178.
145. AbdelMaksoud E, Barakat S, Elmogy M. Diabetic retinopathy grading based on a hybrid deep learning model. 2020 International Conference on Data Analytics for Business and Industry: Way Towards a Sustainable Economy (ICDABI). IEEE. 2020:1–6. https://doi.org/10.1109/ICDABI51230.2020.9325672.
146. Harihanth K, Karthikeyan B. Diabetic retinopathy detection using ensemble deep learning and individual channel training. 2020 3rd International Conference on Intelligent Sustainable Systems (ICISS). IEEE. 2020:1042–49. https://doi.org/10.1109/ICISS49785.2020.9315991.
147. Oulhadj M, Riffi J, Chaimae K, Mahraz AM, Ahmed B, Yahyaouy A, Fouad C, Meriem A, Idriss BA, Tairi H. Diabetic retinopathy prediction based on deep learning and deformable registration. Multimedia Tools Appl. 2022;81(20):28709–27. https://doi.org/10.1007/s11042-022-12968-z.
148. Kale Y, Sharma S. Detection of five severity levels of diabetic retinopathy using ensemble deep learning model. Multimedia Tools Appl. 2023;82(12):19005–20. https://doi.org/10.1007/s11042-022-14277-x.
149. Bodapati JD, Balaji BB. Self-adaptive stacking ensemble approach with attention based deep neural network models for diabetic retinopathy severity prediction. Multimedia Tools Appl. 2024;83(1):1083–102. https://doi.org/10.1007/s11042-023-15120-7.
150. Han J, Ma K-K. Fuzzy color histogram and its use in color image retrieval. IEEE Trans On Image Process. 2002;11(8):944–52. https://doi.org/10.1109/TIP.2002.801585.
151. Sabour S, Frosst N, Hinton GE. Dynamic routing between capsules. Advances in neural information processing systems 2017;30. https://doi.org/10.48550/arXiv.1710.09829.
152. Hinton GE, Krizhevsky A, Wang SD. Transforming auto-encoders. Artificial Neural Networks and Machine Learning–ICANN 2011: 21st International Conference on Artificial Neural Networks. Espoo, Finland, June 14-17, 2011, Proceedings, Part I 21, Springer; 2011:44–51. https://doi.org/10.1007/978-3-642-21735-76.
153. Kumar G, Chatterjee S, Chattopadhyay C. Dristi: A hybrid deep neural network for diabetic retinopathy diagnosis. Signal, Image Video Process. 2021;15(8):1679–86. https://doi.org/10.1007/s11760-021-01904-7.
154. Zago GT, Andreão RV, Dorizzi B, Salles EOT:. Diabetic retinopathy detection using red lesion localization and convolutional neural networks. Comput Biol Med. 2020;116:103537. https://doi.org/10.1016/j.compbiomed.2019.103537.
155. Wan C, Chen Y, Li H, Zheng B, Chen N, Yang W, Wang C, Li Y. Eadnet: A novel lesion segmentation method in diabetic retinopathy using neural networks. Disease Markers. 2021;2021. https://doi.org/10.1155/2021/6482665.
156. Nazir T, Nawaz M, Rashid J, Mahum R, Masood M, Mehmood A, Ali F, Kim J, Kwon H-Y, Hussain A. Detection of diabetic eye disease from retinal images using a deep learning based centernet model. Sensors. 2021;21(16):5283. https://doi.org/10.3390/s21165283.
157. Agrawal V, Kumar V, Sharma S, Chawla R, Paul K. Netradeep: An integrated deep learning and image processing system for precise detection of hard exudates. ACM Trans Comput Healthc. 2024;5(4):1–34. https://doi.org/10.1145/3681796.
158. Akella PL, Kumar R. An advanced deep learning method to detect and classify diabetic retinopathy based on color fundus images. Graefe's Archiv For Clin And Exp Ophthalmol. 2024;262(1):231–47. https://doi.org/10.1007/s00417-023-06181-3.
159. Redmon J, Farhadi A. Yolov3: An incremental improvement. arXiv preprint arXiv:1804.02767. 2018. https://doi.org/10.48550/arXiv.1804.02767.
160. Zhang P, Conze P-H, Lamard M, Quellec G, Daho MEH. Deep learningbased detection of referable diabetic retinopathy and macular edema using ultra-widefield fundus imaging. arXiv preprint arXiv:2409.12854. 2024. https://doi.org/10.48550/arXiv.2409.12854.
161. Gegundez-Arias ME, Marin-Santos D, Perez-Borrero I, Vasallo-Vazquez MJ. A new deep learning method for blood vessel segmentation in retinal images based on convolutional kernels and modified u-net model. Comput Met Programs Biomed. 2021;205:106081. https://doi.org/10.1016/j.cmpb.2021.106081.
162. Jiang Y, Wang F, Gao J, Cao S. Multi-path recurrent u-net segmentation of retinal fundus image. Appl Sci. 2020;10(11):3777. https://doi.org/10.3390/app10113777.
163. Dhal P, Azad C. A novel approach for blood vessel segmentation with exudate detection in diabetic retinopathy. 2020 International Conference on Artificial Intelligence and Signal Processing (AISP). IEEE. 2020:1–6. https://doi.org/10.1109/AISP48273.2020.9073012.
164. Biratu ES, Schwenker F, Ayano YM, Debelee. T.G. A survey of brain tumor segmentation and classification algorithms. J Imag. 2021;7(9):179. https://doi.org/10.3390/jimaging7090179.
165. Bhatkalkar BJ, Nayak SV, Shenoy SV, Arjunan RV. Fundusposnet: A deep learning driven heatmap regression model for the joint localization of optic disc and fovea centers in color fundus images. IEEE Access. 2021;9:159071–80. https://doi.org/10.1109/ACCESS.2021.3127280.
166. Sigut J, Nuñez O, Fumero F, Alayon S, Diaz-Aleman T:. Fovea localization in retinal images using spatial color histograms. Multimedia Tools Appl. 2023;1–19. https://doi.org/10.1007/s11042-023-16244-6.
167. Hervella ÁS, Ramos L, Rouco J, Novo J, Ortega M. Explainable artificial intelligence for the automated assessment of the retinal vascular tortuosity. Med Biol Eng Comput. 2024;62(3):865–81.
168. Badawi SA, Takruri M, Al-Hattab M, Aldoboni G, Guessoum D, ElBadawi I, Aichouni M, Chaudhry IA, Mahar N, Nileshwar AK. Arteriovenous length ratio: A novel method for evaluating retinal vasculature morphology and its diagnostic potential in eye-related diseases. J. Imag. 2023;9(11):253.
169. Badawi SA, Takruri M, ElBadawi I, Chaudhry IA, Mahar NU, Nileshwar AK, Mosalam E. Enhancing vessel segment extraction in retinal fundus images using retinal image analysis and six sigma process capability index. Mathematics. 2023;11(14):3170.
170. Badawi SA, Fraz MM, Shehzad M, Mahmood I, Javed S, Mosalam E, Nileshwar AK. Detection and grading of hypertensive retinopathy using vessels tortuosity and arteriovenous ratio. J Digit Imag. 2022;1–21.
171. Otsu N. A threshold selection method from gray-level histograms. IEEE Trans On Syst, Man, And Cybern. 1979;9(1):62–66. https://doi.org/10.1109/TSMC.1979.4310076.

172. Abdullah Yahya A, Tan J, Su B, Liu K, Hadi AN. Image edge detection method based on anisotropic diffusion and total variation models. J Eng. 2019;2019(2):455–60. https://doi.org/10.1049/joe.2018.5345.
173. Sreedhar K, Panlal B. Enhancement of images using morphological transformation. arXiv preprint arXiv:1203.2514. 2012. https://doi.org/10.48550/arXiv.1203.2514.
174. Heidari AA, Mirjalili S, Faris H, Aljarah I, Mafarja M, Chen H. Harris hawks optimization: Algorithm and applications. Future generation computer systems. 2019;97:849–72. https://doi.org/10.1016/j.future.2019.02.028.
175. Ronneberger O, Fischer P, Brox T. U-net: Convolutional networks for biomedical image segmentation. Medical Image Computing and Computer- Assisted Intervention–MICCAI 2015: 18th International Conference. Munich, Germany, October 5-9, 2015, Proceedings, Part III 18, Springer; 2015 28, pp. 234–41. https://doi.org/10.1007/978-3-319-24574-4.
176. Adarsh R, Amarnageswarao G, Pandeeswari R, Deivalakshmi S. Dense residual convolutional auto encoder for retinal blood vessels segmentation. 2020 6th International Conference on Advanced Computing and Communication Systems (ICACCS). IEEE. 2020:280–84. https://doi.org/10.1109/ICACCS48705.2020.9074172.
177. Zhou Z, Rahman Siddiquee MM, Tajbakhsh N, Liang J. Unet++: A nested u-net architecture for medical image segmentation. Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support: 4th International Workshop, DLMIA 2018, and 8th International Workshop, ML-CDS 2018, Held in Conjunction with MICCAI 2018. Granada, Spain, September 20, 2018, Proceedings 4, Springer. 2018:3–11. https://doi.org/10.1007/978-3-030-00889-5_1.
178. Yan H, Xie J, Zhu D, Jia L, Guo S. Mslf-net: A multi-scale and multi-level feature fusion net for diabetic retinopathy segmentation. Diagnostics. 2022;12(12):2918. https://doi.org/10.3390/diagnostics12122918.
179. Shaukat N, Amin J, Sharif M, Azam F, Kadry S, Krishnamoorthy S. Three-dimensional semantic segmentation of diabetic retinopathy lesions and grading using transfer learning. J Personalized Med. 2022;12(9):1454. https://doi.org/10.3390/jpm12091454.
180. Chen L-C, Zhu Y, Papandreou G, Schroff F, Adam H. Encoder-decoder with atrous separable convolution for semantic image segmentation. Proceedings of the European Conference on Computer Vision (ECCV). 2018, pp. 801–18. https://doi.org/10.1007/978-3-030-01234-2_49.
181. Durai DBJ, Jaya T. Automatic severity grade classification of diabetic retinopathy using deformable ladder bi attention u-net and deep adaptive cnn. Med Biol Eng Comput. 2023;1–23. https://doi.org/10.1007/s11517-023-02860-9.
182. Al-Mukhtar M, Morad AH, Albadri M, Islam MS. Weakly supervised sensitive heatmap framework to classify and localize diabetic retinopathy lesions. Sci Rep. 2021;11(1):23631. https://doi.org/10.1038/s41598-021-02834-7.
183. Sambyal N, Saini P, Syal R, Gupta V. Modified u-net architecture for semantic segmentation of diabetic retinopathy images. Biocybern And Biomed Eng. 2020;40(3):1094–109. https://doi.org/10.1016/j.bbe.2020.05.006.
184. Kandhasamy JP, Balamurali S, Kadry S, Ramasamy LK. Diagnosis of diabetic retinopathy using multi level set segmentation algorithm with feature extraction using svm with selective features. Multimedia Tools Appl. 2020;79:10581–96. https://doi.org/10.1007/s11042-019-7485-8.
185. Tang Y, Pan Z, Pedrycz W, Ren F, Song X. Based kernel fuzzy clustering with weight information granules. IEEE Trans Emerging Top Comput Intel. 2022;7(2):342–56. https://doi.org/10.1109/TETCI.2022.3201620.
186. Pan A, Yang J, Shi C. Research on deep learning-based detection of changes in diabetic retinopathy lesions. Proceedings of the 2023 7th International Conference on Electronic Information Technology and Computer Engineering. 2023, pp. 1702–07. https://doi.org/10.1145/3650400.3650684.
187. Hasan MK, Alam MA, Elahi MTE, Roy S, Martı R:. Drnet: Segmentation and localization of optic disc and fovea from diabetic retinopathy image. Artif Intel Med. 2021;111:102001.
188. Ayano YM, Schwenker F, Dufera BD, Debelee TG. Interpretable machine learning techniques in ecg-based heart disease classification: A systematic review. Diagnostics. 2022;13(1):111. https://doi.org/10.3390/diagnostics13010111.
189. Ayano YM, Schwenker F, Dufera BD, Debelee TG, Ejegu YG. Interpretable hybrid multichannel deep learning model for heart disease classification using 12-leads ecg signal. IEEE Access. 2024. https://doi.org/10.1109/ACCESS.2024.3421641.
190. Shorfuzzaman M, Hossain MS, El Saddik A. An explainable deep learning ensemble model for robust diagnosis of diabetic retinopathy grading. ACM Trans On Multimedia Comput, Commun, And Appl (TOMM). 2021;17(3s):1–24. https://doi.org/10.1145/3469841.
191. Obayya M, Nemri N, Nour MK, Al Duhayyim M, Mohsen H, Rizwanullah M, Sarwar Zamani A, Motwakel A. Explainable artificial intelligence enabled teleophthalmology for diabetic retinopathy grading and classification. Appl Sci. 2022;12(17):8749. https://doi.org/10.3390/app12178749.
192. Singha A, Singhb KK. Fedddr: A federated improved densenet for classification of diabetic retinopathy. Proceedings. 2023. ISSN 1613, 0073. http://ceur-ws.org.
193. Chetoui M, Akhloufi MA. Federated learning for diabetic retinopathy detection using vision transformers. BioMedinformatics. 2023;3(4):948–61. https://doi.org/10.3390/biomedinformatics3040058.
194. Bhulakshmi D, Rajput DS. Privacy-preserving detection and classification of diabetic retinopathy using federated learning with feddeo optimization. Syst Sciamp; Control Eng. 2024;12(1):2436664. https://doi.org/10.1080/21642583.2024.2436664.
195. Raj GM, Morley MG, Eslami M. Federated learning for diabetic retinopathy diagnosis: Enhancing accuracy and generalizability in under-resourced regions. arXiv preprint arXiv:2411.00869. 2024. https://doi.org/10.48550/arXiv.2411.00869.
196. Bhulakshmi D, Rajput DS. Feddl: Personalized federated deep learning for enhanced detection and classification of diabetic retinopathy. PeerJ Comput Sci. 2024;10:2508. https://doi.org/10.7717/peerj-cs.2508.
197. Niu Y, Gu L, Zhao Y, Lu F. Explainable diabetic retinopathy detection and retinal image generation. IEEE J Biomed And Health Inf. 2021;26(1):44–55. https://doi.org/10.1109/JBHI.2021.3110593.
198. Hashim FA, Hussain K, Houssein EH, Mabrouk MS, Al-Atabany W. Archimedes optimization algorithm: A new metaheuristic algorithm for solving optimization problems. Appl Intell. 2021;51:1531–51. https://doi.org/10.1007/s10489-020-01893-z
199. McMahan B, Moore E, Ramage D, Hampson S, Arcas BA. Communication-efficient learning of deep networks from decentralized data. In: Artificial intelligence and statistics. PMLR. 2017:1273–82.
200. Li T, Sahu AK, Talwalkar A, Smith V. Federated learning: Challenges, methods, and future directions. IEEE Signal Process Mag. 2020;37(3):50–60. https://doi.org/10.1109/MSP.2020.2975749.
201. Zhou Y, Chia MA, Wagner SK, Ayhan MS, Williamson DJ, Struyven RR, Liu T, Xu M, Lozano MG, Woodward-Court P, et al. A foundation model for generalizable disease detection from retinal images. Nature. 2023;622(7981):156–63.

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